

CHAPTER 56

Mediastinum

Strictly speaking, the mediastinum is the visceral compartment between the two lungs and includes the mediastinal pleura. However, it is commonly defined as the region between the two pleural sacs, bounded anteriorly by the sternum and posteriorly by the thoracic vertebral column, and extending vertically from the thoracic inlet to the diaphragm. It communicates with the neck via the superior thoracic aperture and with the extraperitoneal, extrapleural and epidural spaces. Beyond each pulmonary hilum, the mediastinum merges into the lung interstitium, incorporating the bronchial tree with its accompanying neurovascular bundles and the pulmonary vasculature.

The posterior mediastinal boundary is longer as a result of the oblique disposition of the superior thoracic aperture and the diaphragmatic curvature posteriorly. With respect to the median sagittal plane, the mediastinum appears asymmetrical.

The mediastinum is divided into superior and inferior mediastina by a plane that crosses the manubriosternal joint and the lower surface of the fourth thoracic vertebra (Fig. 56.1). The inferior mediastinum is subdivided into anterior, middle and posterior parts.

Detailed accounts of mediastinal contents are included with descriptions of the respiratory organs (Ch. 54) and the heart (Ch. 57).

SUBDIVISIONS OF THE MEDIASTINUM

SUPERIOR MEDIASTINUM

The superior mediastinum lies between the manubrium sterni anteriorly and the upper four thoracic vertebrae posteriorly, and is bounded laterally by the mediastinal pleurae (see Fig. 56.1). The reflection of the costomediastinal pleurae follows a line that passes from the sternoclavicular joints in an inferomedial direction, to join the midline on the posterior aspect of the sternal angle. The resultant triangular space is occupied by the thymus or its remnants. The superior mediastinum accommodates the trachea, oesophagus, aortic arch, brachiocephalic

trunk, left common carotid and subclavian arteries (the vertebral artery sometimes arises from the aortic arch between them), the internal thoracic arteries, the left superior intercostal, hemiazygos, internal thoracic and inferior thyroid veins, and numerous lymph nodes, including those from the tracheobronchial, paratracheal and brachiocephalic groups. Laterally, the superior mediastinal structures have the appearance of being 'shrink-wrapped' by the tightly applied pleura. The superior mediastinum transmits the sympathetic and parasympathetic cardiac nerves, the phrenic nerve laterally, the vagus nerves medially and the left recurrent laryngeal nerve. The terminal part of the thoracic duct emerges from the posterior aspect of the aorta and oesophagus, ascending between the left subclavian and carotid arteries. The left and right brachiocephalic veins form the superior vena cava. The superior (extrapericardial) half of the superior vena cava and its major tributaries are situated on a more superficial plane posterior to the manubrium and the origins of sternothyroid inferiorly, and sternohyoid superiorly. An enlarged thyroid gland may extend inferiorly between the sternum and the brachiocephalic veins; its surgical excision requires due diligence from possible venous adherence. The origins of longus colli (inferior oblique and vertical intermediate portions) are the deepest elements of the superior mediastinum.

INFERIOR MEDIASTINUM

Anterior mediastinum

The anterior mediastinum lies between the sternal body and pericardium (see Fig. 56.1). It narrows above the fourth costal cartilages where the pleural sacs converge, and contains loose connective tissue, the sternopericardial ligaments, a few lymph nodes, the mediastinal branches of the internal thoracic artery, and sometimes part of the thymus gland or its degenerated remains. The paired intervals between the sternal and costal muscular slips of the diaphragm are filled with areolar tissue and are crossed by the superior epigastric vessels and by some lymph vessels from the abdominal wall and liver; they are rare sites of abdominal herniation. The pericardium and the heart are routinely approached by either a complete median or partial sternotomy. The sternopericardial ligaments and the pleural reflections are easily separated by blunt dissection. In case of pericardial tamponade, the pericardial cavity can be easily drained through a subxyphoid approach, either surgically or via needle pericardiocentesis.

Middle mediastinum

The middle mediastinum is the broadest part of the inferior mediastinum (see Fig. 56.1) and contains the pericardium, heart and ascending aorta, the inferior, intrapericardial half of the superior vena cava (receiving the azygos venous arch posteriorly), the tracheal bifurcation and main bronchi, the pulmonary trunk, right and left pulmonary vasculature, phrenic nerves, the deep part of the cardiac plexus and the tracheobronchial lymph nodes. The short thoracic part of the inferior vena cava, both extra- and intrapericardial segments, extends between the vena caval aperture of the diaphragm and its termination in the right atrium. The structures forming the pulmonary root converge at the level of the hilum. The mediastinal pleura is continuous with the visceral pleura at the level of the hilum, which is where the lateral boundary of the middle mediastinum is conventionally located. The fibrous pericardium lies on, and is fused with, the anterior two-thirds of the central tendon of the diaphragm.

Posterior mediastinum

The posterior mediastinum is the longest and narrowest portion of the inferior mediastinum (see Fig. 56.1). It is bounded anteriorly by the

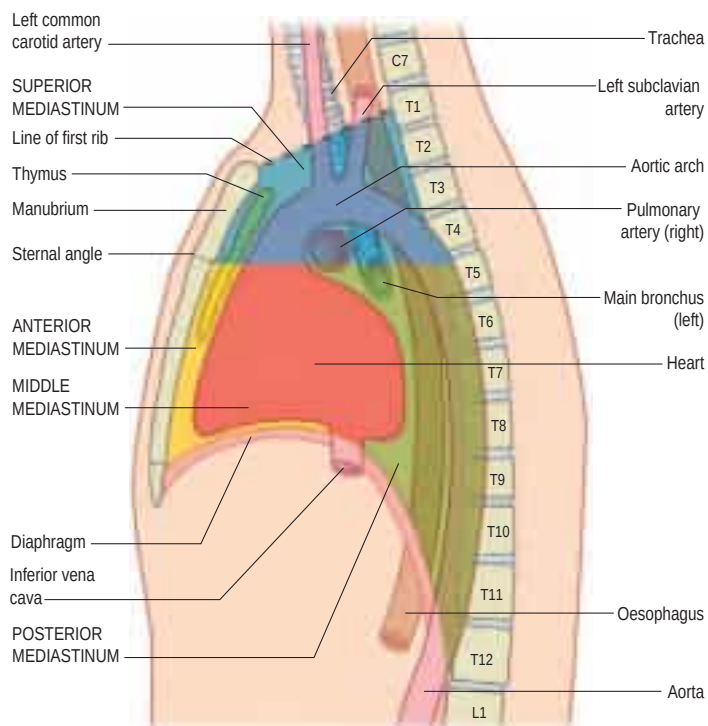


Fig. 56.1 The mediastinal divisions.

tracheal bifurcation, pericardium and pulmonary vessels, and posteriorly by the bodies of the fifth to the twelfth thoracic vertebrae. There is no strict demarcation with the superior mediastinum; numerous structures travel through both regions. The arched posterior third of the central portion of the diaphragm constitutes the anteroinferior limit of the posterior mediastinum laterally as the mediastinal pleurae come close together. Two pleural recesses, the interaortico-oesophageal and interazygo-oesophageal recesses, intercalate from the left and right sides between the aorta and oesophagus and the azygos vein and oesophagus, respectively.

The posterior mediastinum contains the descending thoracic aorta (on the left side of the spine), the oesophagus (median, but positioned anterior to the aorta inferiorly) and, more posteriorly, the azygos and hemiazygos venous systems, the thoracic duct, lymph nodes, right and left sympathetic chains and thoracic splanchnic nerves. The vagal trunks course adjacent to the oesophagus; the anterior trunk is constituted mainly from the left vagus nerve and the posterior mainly from the right vagus nerve. The descending thoracic aorta gives off the intercostal arteries and the largest anterior segmental medullary artery, the 'great radicular artery of Adamkiewicz' (Yoshioka et al 2003) (p. 770).

There are a number of communications between the posterior mediastinum and the abdomen. These are: the oesophageal aperture (also transmits the vagal trunks) at the level of the tenth thoracic vertebra; the aortic aperture (also transmits the thoracic duct and occasionally the azygos vein) at the level of the twelfth thoracic vertebra; apertures within the diaphragmatic crura that transmit the thoracic splanchnic nerves; apertures deep to the medial lumbosacral arches for the sympathetic chains; and minute openings in the central tendon of the diaphragm that transmit small veins. All openings represent potential communication sites for suppurative or neoplastic processes.

MEDIASTINAL COMMUNICATIONS WITH THE NECK

Anatomical pathways exist between the oral cavity and the thorax via the parapharyngeal space and along other fascial planes of the neck. The former is more likely to be infected than any of the other potential tissue spaces in the head and neck, communicating with the retropharyngeal and pretracheal spaces, so reaching the superior mediastinum and then the anterior part of the inferior mediastinum (see Chs 29 and 31). The plane between the buccopharyngeal and prevertebral fasciae is a highway for spread of air and gastric contents between the neck and mediastinum after oesophageal injury (Wind and Valentine 2013). The carotid sheath, containing the carotid arteries, jugular veins and vagus nerves, represents another potential route of communication.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

The mediastinum contains the great vessels (ascending aorta, aortic arch and its branches, descending thoracic aorta, superior vena cava and the terminal part of the inferior vena cava), the pulmonary arteries and veins, the internal thoracic and posterior intercostal arteries and veins, the azygos and hemiazygos venous systems, and the thoracic duct.

Great vessels of the superior mediastinum

The aortic arch, descending thoracic aorta, pulmonary trunk and superior vena cava are described in Chapter 57.

Azygos venous system

Azygos vein

The azygos vein has a variable origin from the azygos lumbar vein or from the posterior aspect of the inferior vena cava at or below the level of the renal veins (Figs 56.2–56.3; see Fig. 56.14B). When present, the azygos lumbar vein ascends anterior to the upper lumbar vertebral bodies and passes either posterior to or through the right diaphragmatic crus or traverses the aortic hiatus to the right of the cisterna chyli. Anterior to the twelfth thoracic vertebra, it is joined by a venous trunk, formed by the union of the right ascending lumbar and subcostal veins, to form the azygos vein. In the absence of a lumbar azygos vein, this common trunk may continue as the azygos vein proper. The azygos vein passes to the right, posterior to the right crus of the diaphragm, and ascends in the posterior mediastinum to the level of the fourth thoracic vertebra, where it arches anteriorly, superior to the right pulmonary hilum, and joins the posterior aspect of the superior vena cava, just superior to its pericardial incorporation.

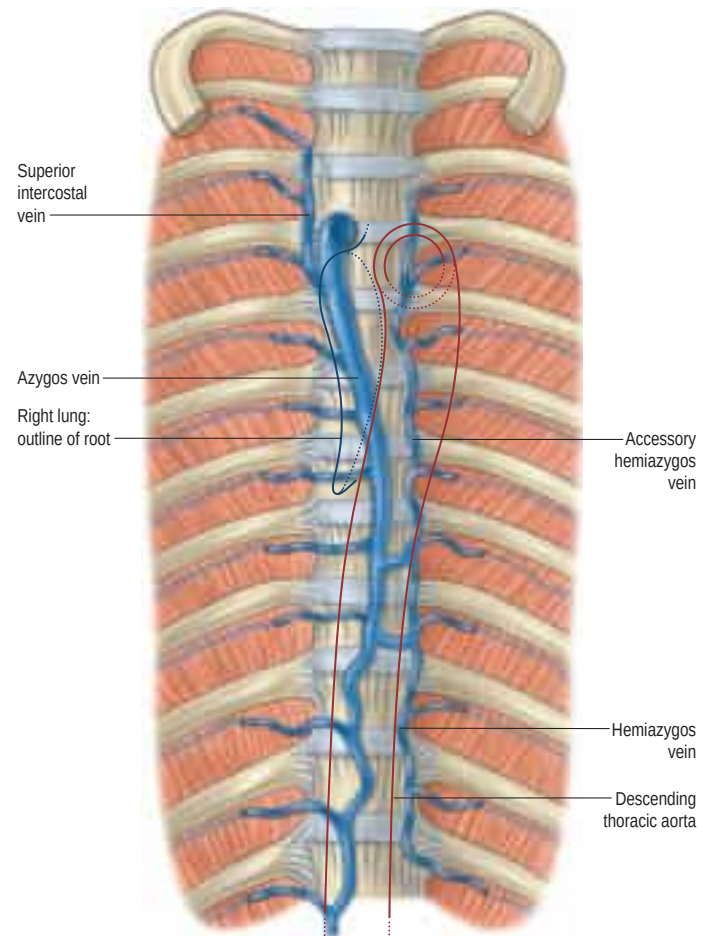


Fig. 56.2 A common course followed by the intrathoracic azygos, hemiazygos and accessory azygos veins. Outlines of the root of the right lung and ascending thoracic aorta are included.

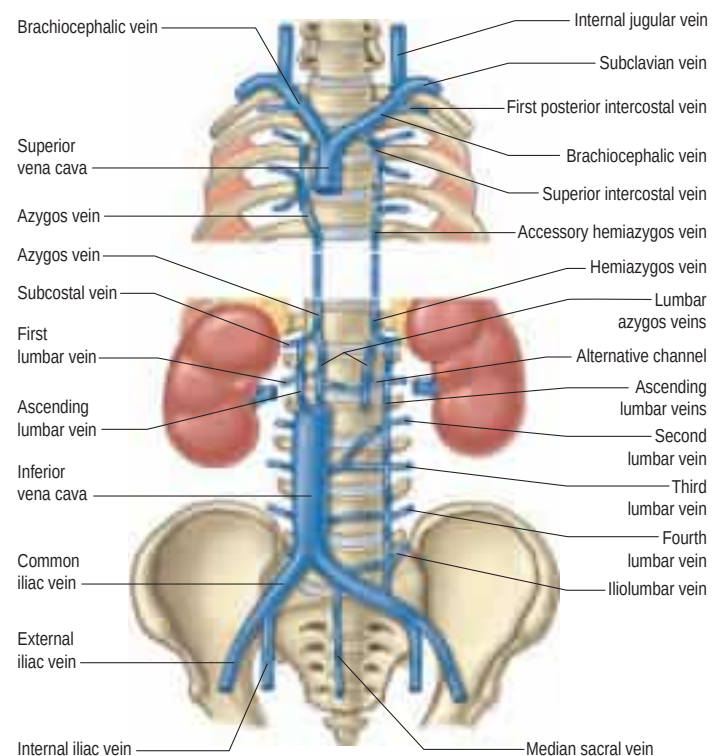


Fig. 56.3 The most common topography of the azygos venous system.

In its course, the azygos vein lies anterior to the bodies of the lower eight thoracic vertebrae, anterior longitudinal ligament and right posterior intercostal arteries. Right lateral relations are the right sympathetic chain, the right greater splanchnic nerve, lung and pleura. Left lateral relations are the thoracic duct, aorta and, where the vein arches anteriorly, the oesophagus, trachea and right vagus. Aortic pulsations

may assist venous return in the azygos and hemiazygos veins, although only a small and variable number of imperfect valves are present. Inferiorly, the azygos vein lies posterior to the oesophagus, from which it is separated by a pleural recess, the azygo-oesophageal recess. There are differences in the appearance of the contour of this recess on computed tomography (CT) between children and adults. The recess is dextroconvex below the age of 6 years, non-concave (equally divided between straight and convex) between 6 and 12 years and concave, as in adults, after the age of 12 years (Miller et al 1993). Its major tributaries are the right intercostal veins (the first drains into the right brachiocephalic vein and the second, third and fourth form a common trunk, the superior intercostal vein), the right ascending lumbar and subcostal veins, mediastinal, oesophageal, pericardial and right bronchial veins, and the hemiazygos and accessory hemiazygos veins.

Numerous anastomoses exist between the tributaries and the main trunks of the azygos, hemiazygos and accessory hemiazygos veins, the inferior vena cava and the vertebral venous plexuses. These allow for alternate drainage pathways in case of segmental agenesis ('interruption') or thrombosis of the inferior vena cava (see also below for the variations of the azygos vein).

Hemiazygos vein

The hemiazygos vein represents the left-sided equivalent of the more inferior part of the azygos vein. It originates in a similar fashion (from the union of the left lumbar ascending and subcostal veins) but exhibits more variability; it may also connect with the left renal vein via the equivalent of the lumbar azygos vein. The hemiazygos vein crosses posterior to the aorta and oesophagus, and either anterior or posterior to the thoracic duct, and usually terminates in the azygos vein at the level of the eighth thoracic vertebra (see Fig. 56.2). It receives the lower three left posterior intercostal, mediastinal and oesophageal veins.

Accessory hemiazygos vein

The accessory hemiazygos vein functions as the left-sided mirror image of the superior portion of the azygos vein. It drains the fifth to eighth posterior intercostal veins and occasionally the left bronchial veins, and usually terminates in the azygos vein after crossing the seventh thoracic vertebra. The first posterior intercostal vein drains directly into the left brachiocephalic vein, while the second to the fourth drain indirectly to the accessory hemiazygos vein via the left superior intercostal vein. The accessory hemiazygos vein may also join the hemiazygos vein. One or more prevertebral transversal anastomoses may be present.

Variations of the azygos venous system

The origin, course, tributaries, anastomoses and termination of the azygos system of veins are all highly variable. The arrangement shown in Figure 56.2 represents the most common pattern with a right-sided azygos vein. The left channel may be absent or underdeveloped, allowing direct drainage of the left intercostal veins into the azygos and/or the termination of the accessory hemiazygos into the left brachiocephalic vein. In such cases, the azygos vein is in median position.

Internal thoracic arteries and veins

The internal thoracic vessels are described on pages 942 and 943.

Pulmonary arteries and veins

The pulmonary vessels are described on pages 959 and 960.

Posterior intercostal arteries and veins

The posterior intercostal veins accompany their arteries in eleven pairs; they are described on page 944.

Mediastinal lymph nodes

For the purposes of staging lung cancer, thoracic surgeons classify the mediastinal lymph nodes into regional lymph node stations, which are defined as shown in Table 56.1 and Figure 56.4.

These groups are not sharply demarcated. Pulmonary nodes become continuous with the bronchopulmonary nodes, and these, in turn, merge with the inferior and superior tracheobronchial nodes, which are continuous with the paratracheal group. The tracheobronchial nodes drain the lungs, bronchi, the thoracic part of the trachea, and heart, and also receive efferents from the posterior mediastinal nodes. Their efferent vessels ascend on the trachea to unite with efferents of the parasternal

Table 56.1 Classification of regional mediastinal lymph nodes into regional lymph node stations

Station	Description
1	The highest mediastinal nodes lying above a horizontal line, at which level the left brachiocephalic vein crosses the trachea
2	Upper paratracheal nodes lying below the line of the highest mediastinal nodes and above a line drawn horizontally at the level of the upper border of the aortic arch
3	Prevascular and retrotracheal nodes lying behind the trachea but in front of the great vessels
4	Lower paratracheal nodes lying below the upper margin of the aortic arch and down to the upper margin of the corresponding upper lobe bronchus. On the right side, this is the upper margin of the right upper lobe bronchus; the majority of nodes in this area tend to be positioned anterolateral to the trachea. On the left side, the nodes are located below the upper margin of the aortic arch and above the margin of the left upper lobe bronchus. They lie to the right of the ligamentum arteriosum, between the ligament and the trachea
5	Subaortic nodes lying in the aortopulmonary window and situated to the right of the ligamentum arteriosum, aorta or left pulmonary artery, but proximal to the first division of the left pulmonary artery
6	Para-aortic nodes lying between the upper margin of the aortic arch and lateral to the ascending aorta and aortic arch
7	Subcarinal nodes are not associated with the lower lobe bronchi
8	Para-oesophageal nodes lie well below the subcarinal nodes, close to the pulmonary veins
9	Nodes lying within the pulmonary ligament
10, 11 and 12	Designated the hilar, interlobar and lobar nodes, respectively

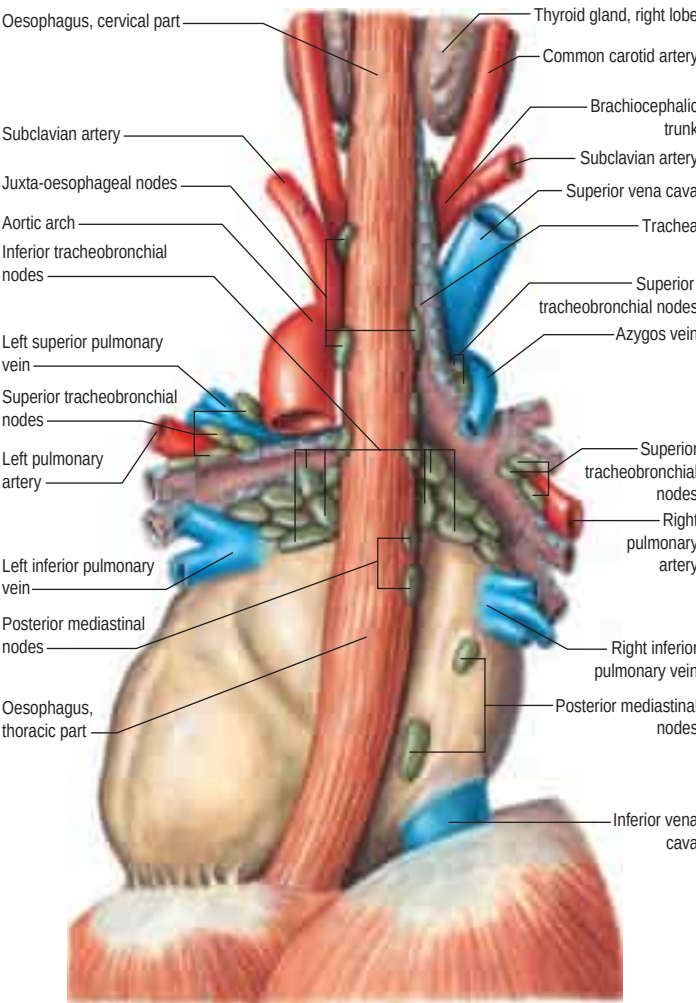


Fig. 56.4 The general disposition of the thoracic lymph nodes, posterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

nal and brachiocephalic nodes as the right and left bronchomediastinal trunks. The right trunk may occasionally join a right lymphatic duct or another right-sided lymph trunk, and the left trunk may join the thoracic duct, but usually they open independently in or near the ipsilateral jugulosubclavian junction (see Fig. 29.16).

In congenital interruption of the inferior vena cava or acquired thrombosis, the azygos system provides an alternative pathway for venous drainage. A higher than usual azygos arch may travel within an accessory azygos fissure (four pleural layers) and isolate a portion of the right upper lobe of the lung medially: the 'azygos lobe', which may rarely be radiopaque on a chest radiograph. An azygos lobe is rarely present and is rarely seen on chest radiographs. The azygos fissure often presents a nestled azygos vein as a lung 'pseudo-nodule' on a chest radiograph; it is typically comma-shaped and should not to be confused with a more rounded early lung carcinoma or metastasis.

Metastatic involvement of these lymph nodes has important prognostic implications and influences the choice of treatment ([Mountain and Dresler 1997](#)). The staging system for lung cancer classifies involvement of ipsilateral hilar lymph nodes as N1, ipsilateral mediastinal lymph nodes as N2, contralateral mediastinal or hilar nodes as N3 and supraclavicular/scalene nodes as M1 (distant metastases).

Mediastinoscopy

The decision to proceed with mediastinoscopic biopsy depends on whether suitable nodal stations seem involved on CT scanning and whether there is a history of prior neck and mediastinal surgical procedures.

THORACIC DUCT

The thoracic duct conveys the lymph from the entire body, except the right part of the head, neck and heart, right upper limb, right lung and part of the convex surface of the liver, back to the venous circulation. It usually originates anterior to the first or second lumbar vertebrae, where the intestinal and two lumbar lymph trunks join to form the cisterna chyli. (Both the thoracic duct and the cisterna chyli are variable in their origin, course and length; the latter is even more variable in width and shape than the thoracic duct.) From the superior aspect of the cisterna chyli, the thoracic duct passes through the aortic hiatus and enters the posterior mediastinum, lying to the right between the aorta and the azygos vein up to the level of the fifth thoracic vertebra, where it typically crosses over the vertebral column posterior to the oesophagus (see Fig. 56.7B). It continues upwards, first posterior to the aortic arch adjacent to the left side of the oesophagus, then posterior to the left subclavian artery in the superior mediastinum. It next arches over the subclavian artery and descends anteriorly to empty into the venous circulation in the region of the left jugular and subclavian veins, either at their very confluence or in proximity, sometimes in the left brachiocephalic vein (Fig. 56.5, p. 462). In the space between longus colli, the anterior scalene muscles and the pleural dome, the duct usually arches

3 or 4 cm superior to the clavicle, crossing anterior to the vertebral vessels, the left sympathetic trunk, the vertebral ganglion, the thyrocervical trunk and the left phrenic nerve.

In some patients, two thoracic ducts exist within the mediastinum, and occasionally a single thoracic duct may empty into right-sided venous structures, the latter disposition being more frequent in individuals with an aberrant right subclavian artery (the latter originating from the distal aortic arch ascending posterior to the oesophagus).

When present as a single channel, the thoracic duct is 5 mm in diameter at its origin, narrows to 2–3 mm at mid-thoracic level and then widens again at its termination. Alternatively, it may divide into two or three branches that eventually reunite or drain into both the left and right venous systems. The duct appears sinuous and has valves throughout its length; a bicuspid valve at its termination may prevent the backflow of blood. As it passes through the mediastinum, the thoracic duct also receives non-chylous lymph from tributaries that drain regions of the pulmonary parenchyma and parietal pleura. The sum of these sources accounts for a total lymphatic flow through the thoracic duct of 1500–2400 ml per day in proportion to the dietary intake of fat, particularly long-chain triglycerides.

The anatomy of the thoracic duct determines the location of the effusion seen with duct injury or obstruction.

Tributaries of the thoracic duct

Some tributaries of the thoracic duct may terminate directly in the venous system. Numerous anastomoses exist between visceral, parietal, limb and mammary lymphatics, an arrangement that is manifest in observed multiple patterns of dissemination of cancer cells. Besides the cisterna chyli and its tributaries, bilateral ascending lumbar lymph trunks from the upper lateral aortic nodes pierce the corresponding diaphragmatic crus and eventually join the thoracic duct in the posterior mediastinum. The inferior six or seven intercostal lymph trunks from the corresponding bilateral intercostal nodes descend to terminate in the duct, either directly or by variable common trunks. The left upper intercostal lymphatics may join the cervical portion of the duct or the subclavian trunk. The left subclavian trunk usually joins the thoracic duct but may open independently into the left subclavian vein. These mediastinal trunks provide lymph drainage paths to the thoracic duct from the convex diaphragmatic aspect of the liver, the diaphragm, the pericardium, heart and oesophagus, and many possess terminal bicuspid valves. The left bronchomediastinal trunk occasionally joins the thoracic duct but usually has an independent venous termination. The left jugular trunk usually joins the thoracic duct but may open independently into the left internal jugular vein.

Injury during oesophageal surgery

The variable origin, course and length of the thoracic duct, coupled with failure to identify it at surgery, may lead to its inadvertent incision or transection, particularly during trans-hiatal and thoracoscopic oesophageal surgery, with an incidence between 0.2 and 3% (Wemyss-Holden et al 2001). Thoracic duct laceration is a potentially life-threatening complication; mortality rates are higher with conservative management but still elevated even after surgical ligation. Rupture of the thoracic duct leads to leakage of chyle, which is rich in lipid, protein, T lymphocytes (ranging from 400 to 6800 cells/ μ l), immunoglobulins and fat-soluble vitamins, and hence a progressive nutritional and immune deficit develops. Postoperative chylous effusions are usually the result of damage to ductal tributaries rather than to the duct itself, and are usually self-limiting. The greater incidence of injury with trans-hiatal resection may be attributable to shear forces during the mobilization of the distal oesophagus, whereas the limited field of view probably contributes to the greater incidence of thoracic duct injury during thoracoscopic surgery. Injury should be suspected in the postoperative period if there is either an enlarging mediastinal silhouette on serial chest radiographs or a significant milky drainage from the chest or abdominal drains. An electrophoretic confirmation of the presence of chylomicrons in the pleural fluid is diagnostic, amongst other tests. The thoracic duct may be injured during the left cervical approach for the exposure of the vertebral and subclavian arteries or the cervicothoracic (stellate) sympathetic ganglion. However, deliberate ligation of the duct at this level represents an accepted surgical manoeuvre with no particular consequences.

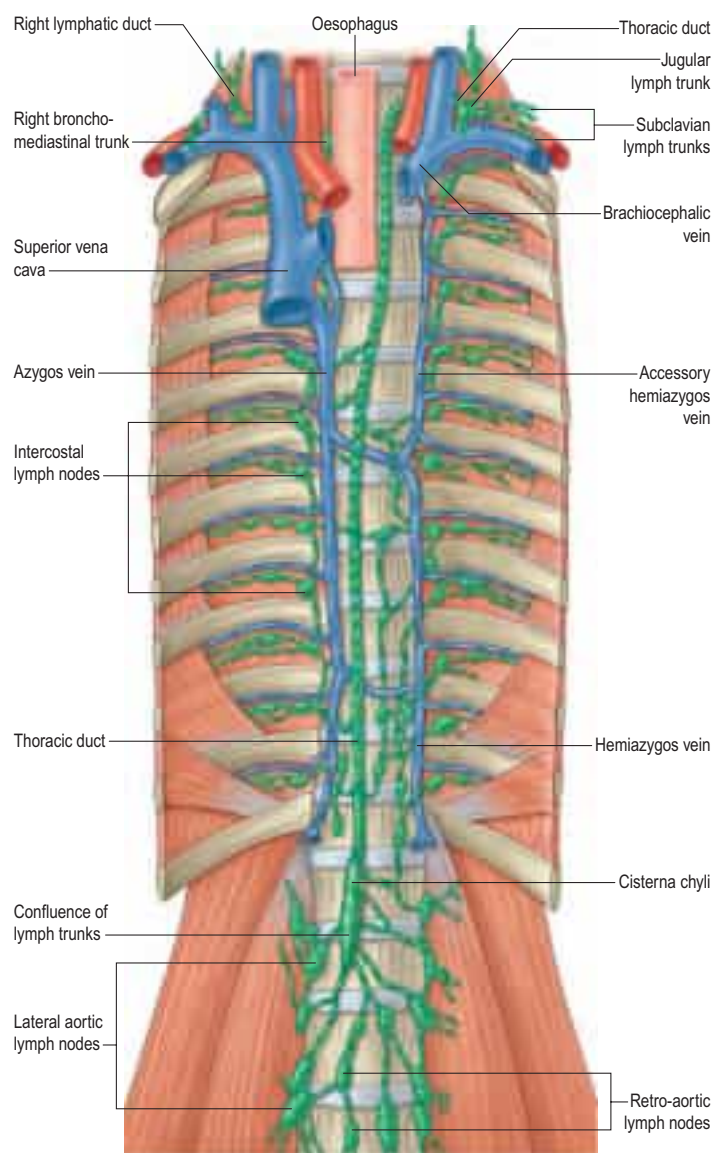


Fig. 56.5 The thoracic and right lymphatic ducts. Note that the accessory hemiazygos vein is crossing the median plane lower and the hemiazygos higher than typically found.

RIGHT LYMPHATIC TRUNK

The right lymphatic trunk has an inconstant anatomy (see Fig. 56.5). Its three main tributaries – the right jugular, subclavian and

Mediastinoscopy can be performed with video-assisted or direct optic visualization. A transverse incision is made in the suprasternal notch through the deep lamina of the deep cervical fascia after retraction of sternohyoid and sternothyroid within its superficial lamina. The pretracheal plane is dissected and a space is created between the anterior aspect of the trachea and the brachiocephalic trunk inferior to the carina and along both sides of the trachea and main bronchi. This space facilitates biopsy of the upper paratracheal 2R/L stations, the right paratracheal 4R station and higher nodes from the left paratracheal 4L station superior to the aortic arch. Anterior subcarinal station 7 and bilateral hilar stations 10R/L may be accessible but are technically challenging. Nodes can be sampled or removed entirely for histological evaluation.

The anterior part of the superior mediastinum is more difficult to assess by mediastinoscopy and usually requires an anterior mediastinotomy (Chamberlain procedure). This is performed to biopsy nodes from the more inferior aspect of the left paratracheal 4L station, sub-aortic station 5, para-aortic station 6 and subcarinal station 7, and also facilitates anterior mediastinal mass biopsy.

Complications of mediastinoscopy include haemorrhage, injury to the trachea or left recurrent laryngeal nerve, and pneumothorax.

Endobronchial ultrasound can be performed to biopsy lymph nodes in stations 1–4, 7 and 10–12, using fine needle aspiration through the tracheal or bronchial walls. In a similar way, transoesophageal ultrasound may biopsy subcarinal station 7, para-oesophageal station 8 and pulmonary ligament station 9 nodes.

Because the thoracic duct crosses the mediastinum at, or at about, the level of the fifth thoracic vertebral body, lymphatic injury or obstruction above or below this level results in a left- or right-sided pleural effusion, respectively ([Doerr et al 2001](#)). Anomalies and sites of injury to the duct are identified by lymphangiography, with or without post-procedure CT scanning, or lymphoscintigraphy. Lymphangiography is performed using bilateral localization of pedal lymphatics with a subcutaneous injection of methylene blue, followed by a minor surgical procedure to cannulate the lymphatic vessel, and injection of Lipiodol contrast agent. Conventional plain films and CT images (without intravenous contrast) are then obtained. Lymphangiography can identify areas of chyle leakage and lymphangiectasia in the majority of patients ([Deso et al 2012](#)).

bronchomediastinal trunks – often terminate separately at the jugular and subclavian veins (see Fig. 29.16). In up to one-fifth of individuals, they may aggregate to form the right lymphatic trunk, usually 0.5–1 cm in length. The terminal portions of the right lymphatic trunk and of the thoracic duct are concealed by the scalene fat pad, which renders them more vulnerable during surgery.

AUTONOMIC NERVOUS SYSTEM

The thoracic autonomic nervous system consists of the right and left ganglionated sympathetic chains and vagi, and the cardiac, pulmonary and oesophageal plexuses. The cardiac plexus is described on page 1021, the pulmonary plexus on page 961, and the oesophageal plexus is described below.

THORACIC SYMPATHETIC TRUNK

The thoracic sympathetic trunk contains ganglia that are almost equal in number to those of the thoracic spinal nerves (usually 11, occasionally 12, rarely 10 or 13) (Figs 56.6–56.7). Almost always, the first thoracic ganglion is fused with the inferior cervical ganglion, forming the cervicothoracic (stellate) ganglion; occasionally, the second thoracic ganglion is included in this fusion. The succeeding ganglia are counted in order to make them correspond numerically with the correlative segmental structures. Except for the second and lowest two or three, the thoracic ganglia lie against the costal heads, posterior to the costal pleura. The second thoracic sympathetic ganglion is commonly located in the second intercostal space, and the lowest two or three ganglia lie lateral to the bodies of the corresponding vertebrae. Inferiorly, the thoracic sympathetic trunk passes posterior to the medial arcuate ligament

or through the diaphragmatic crus to become the lumbar sympathetic trunk.

The ganglia are small and interconnected by intervening segments of the trunk. Unlike other mammalian species, including primates, each sympathetic ganglion in humans is connected to two consecutive spinal nerves: the corresponding and succeeding nerves. The classic description is that two or more rami communicantes, named white (myelinated, preganglionic) and grey (unmyelinated, postganglionic), connect each ganglion with its corresponding spinal nerve, the white rami joining the nerve distal to the grey.

Variation in this pattern is common, especially at the upper thoracic levels, but symmetry is rarely seen. Additional ascending and descending rami issue from the second (54% and 46%, respectively), third (6% and 25%) and fourth ganglia (5% and 8%) (Cho 2005). Occasionally, a grey and a white ramus fuse to form a 'mixed' ramus. An inconstant intrathoracic ramus may join the second intercostal nerve to the ventral ramus of the first thoracic nerve proximal to the point where the latter issues a large branch to the brachial plexus (Kuntz 1927). This intrathoracic nerve of Kuntz is considered to carry sympathetic fibres to the brachial plexus without passing through the sympathetic trunk and has been held responsible for the recurrence of symptoms following sympathectomy. The anatomical variation exhibited by the communicating rami and the location of the second sympathetic ganglion may also contribute to surgical failure and symptom recurrence.

The medial branches from the upper five ganglia are very small, and may supply filaments in a retrograde manner to the thoracic aorta and its branches, as they pass to the cardiac plexuses (Kawashima 2011). Some fibres form a fine thoracic aortic plexus with filaments from the greater splanchnic and vagus nerves. Rami of the second to fifth or sixth ganglia enter the posterior pulmonary plexus. Small branches of these pulmonary and cardiac nerves pass to the oesophagus and trachea.

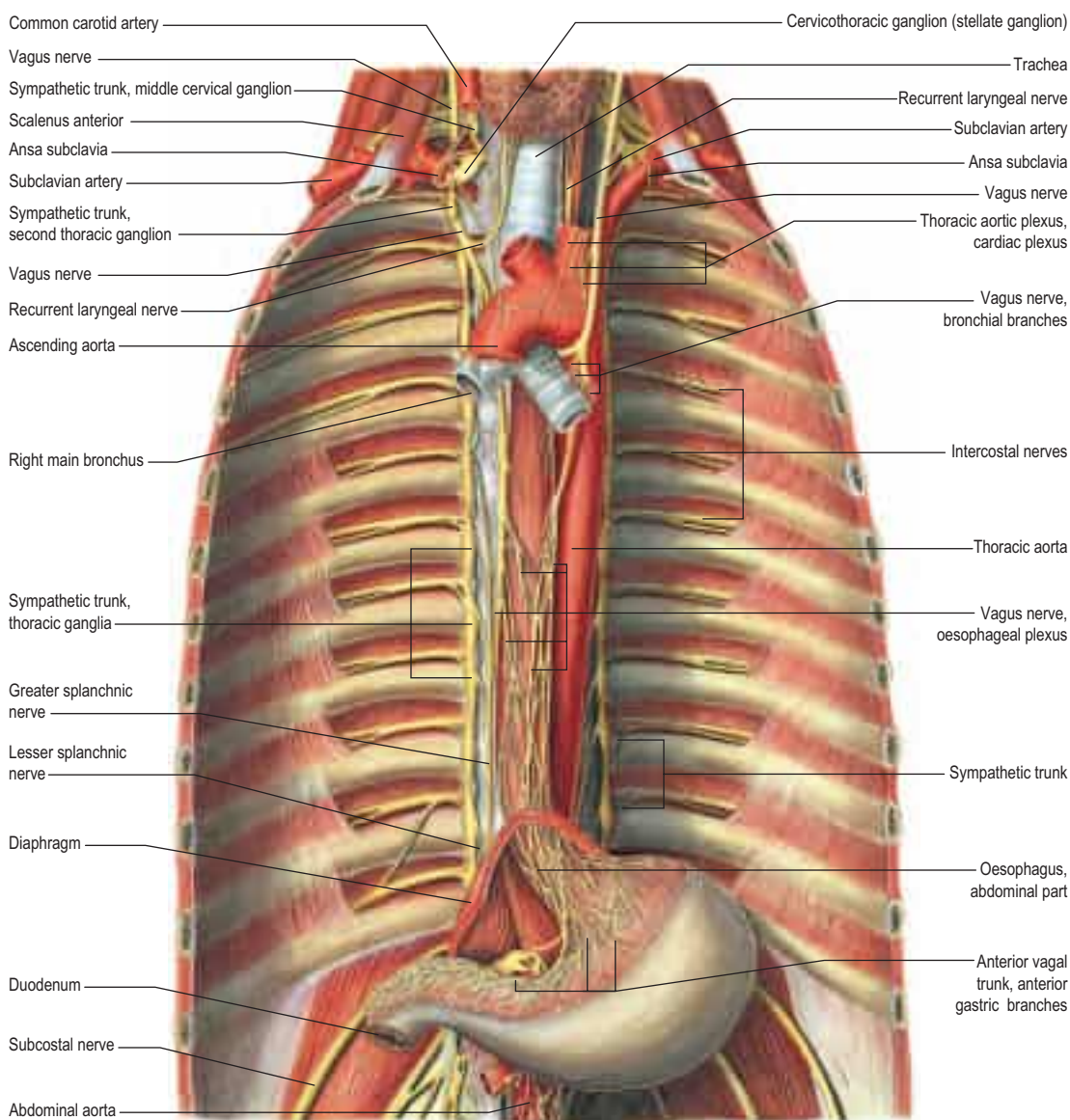


Fig. 56.6 The thoracic autonomic nervous system. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

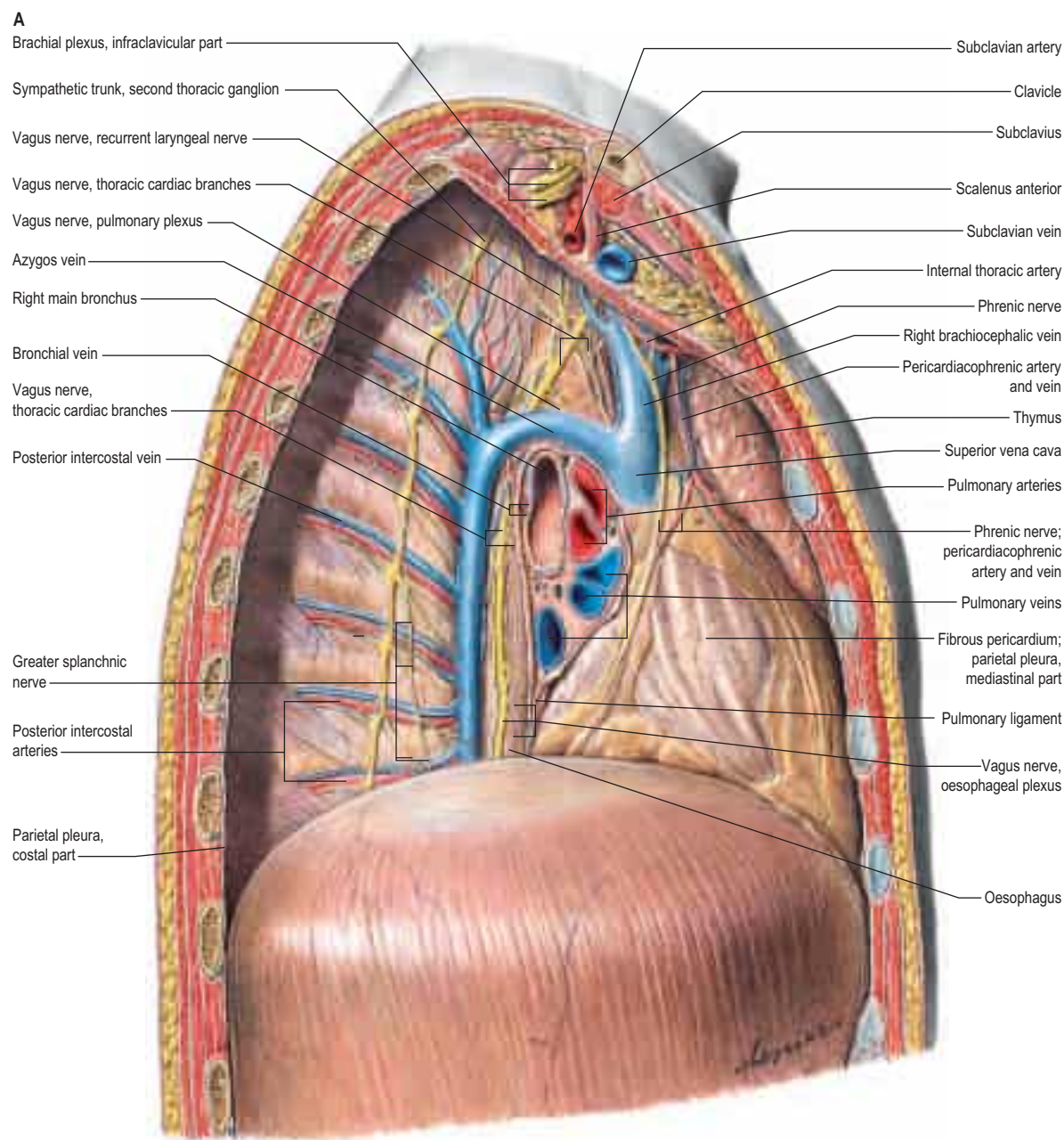


Fig. 56.7 The mediastinum. **A**, Right lateral aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Continued

The larger medial branches from the lower seven ganglia supply the aorta and then unite to form the greater, lesser and least splanchnic nerves (see below for variability).

The greater splanchnic nerve consists mainly of myelinated preganglionic efferent and visceral afferent fibres, and is formed by branches from the fifth to ninth or tenth thoracic ganglia; fibres in the upper branches may be traced to the first or second thoracic ganglia. Its roots vary from one to eight, four being the most usual number. It descends obliquely on the vertebral bodies, supplies branches to the descending thoracic aorta, and perforates the ipsilateral diaphragmatic crus to end mainly in the coeliac but partly in the aorticorenal ganglion and suprarenal (adrenal) gland. A splanchnic ganglion exists on the nerve opposite the eleventh or twelfth thoracic vertebra in a majority of individuals.

The lesser splanchnic nerve, formed by rami of the ninth and tenth (sometimes the tenth and eleventh) thoracic ganglia and the trunk between them, pierces the diaphragm with the greater splanchnic nerve and joins the aorticorenal ganglion.

The lowest (least) splanchnic nerve (or renal nerve) from the lowest thoracic ganglion enters the abdomen with the sympathetic trunk to end in the renal plexus.

The greater splanchnic nerve is always present, the lesser is usually present, and the least is often absent. A fourth (accessory) splanchnic nerve has been described.

Thoracic sympathectomy

Endoscopic thoracic sympathectomy (ETS) is a surgical procedure used to relieve the symptoms of craniofacial, palmar or axillary hyperhidrosis, facial blushing, Raynaud's disease with digital ulcers, and reflex sympathetic dystrophy (shoulder-hand syndrome).

VAGUS NERVE IN THE MEDIASTINUM

The vagus nerve contains preganglionic parasympathetic fibres that arise in its dorsal nucleus and travel in the nerve and in its pulmonary, cardiac, oesophageal, gastric and intestinal branches (see Fig. 56.6). Vagal fibres that originate from neurones in the nucleus ambiguus are distributed to the constrictor muscles of the pharynx and the intrinsic muscles of the larynx. The proportion of efferent parasympathetic fibres in the vagus varies at different levels but is small relative to its sensory and sensorimotor content. Efferent fibres relay in minute ganglia in the visceral walls.

Cardiac branches join the cardiac plexuses and relay in ganglia that are distributed freely over both atria in the subepicardial tissue. The terminal fibres are distributed to the atria and the atrioventricular bundle; they are concentrated around the sinu-atrial and, to a lesser

ETS may be beneficial to ameliorate cerebral vascularization in patients with moyamoya disease and to treat migraine, hyperactive bronchial states, long QT syndrome, causalgic pain, erythromelalgia, Buerger's disease, Prinzmetal's angina, and chronic non-infectious rhinitis. ETS is also effective in frostbite injury, especially if performed within 36–72 hours of cold exposure.

The operation involves making tiny incisions posterior to the pectoral fold in the axilla, and insufflating a small amount of carbon dioxide into the thoracic cavity to allow access with a modified thoracoscope. In the treatment of facial blushing, it is sufficient to divide the fibres that run superiorly from the second thoracic ganglion over the neck of the second rib, leaving the second ganglion almost intact. Treatment of palmar hyperhidrosis requires ablation thermocoagulation of the sympathetic trunk over the necks of the third and fourth ribs, taking care to avoid any spreading of thermal energy along the trunk in order to avoid damaging the more superior stellate ganglion (see 'Horner's syndrome' below). The risk of compensatory sweating is greatly reduced, though not completely excluded, by limiting the number of ganglia treated to an absolute minimum. As an additional procedure, dividing the inconstant sympathetic pathways (nerve of Kuntz, ascending or descending additional rami communicantes) on the second, third and fourth ribs may improve surgical outcomes.

The effect is immediately evident: the patient awakes from anaesthesia with dry, warm hands. In many cases, even hyperhidrosis of the feet improves, but the underlying anatomical/physiological mechanisms are not yet properly understood ([Gofeld and Faclier 2006](#)). Surgical complications are very rare; Horner's syndrome is the most feared, and is caused by damage to the stellate ganglion and interruption of the sympathetic fibres from C8–T1, which ascend around the arteries supplying the head and neck (see pp. 468, 469 and [Fig. 29.18](#)). Side-effects include compensatory sweating (ranging from barely noticeable to quite disturbing) in other locations of the body on exercise or from exposure to high temperatures in up to 70% of patients. Severe in 5% of patients, it may be more frequent in those with axillary hyperhidrosis (when two more inferior ganglia have to be divided); some surgeons do not now consider isolated axillary hyperhidrosis as an indication for this procedure. Gustatory or olfactory sweating may also occur in up to one-third of patients but is rarely considered a problem. Other documented side-effects are the inability to raise the heart rate when working out physically; in some cases, this has led to decreased ability to perform work and daily activities. Some patients also experience an uncomfortable sensation of not being able to control their body temperature.

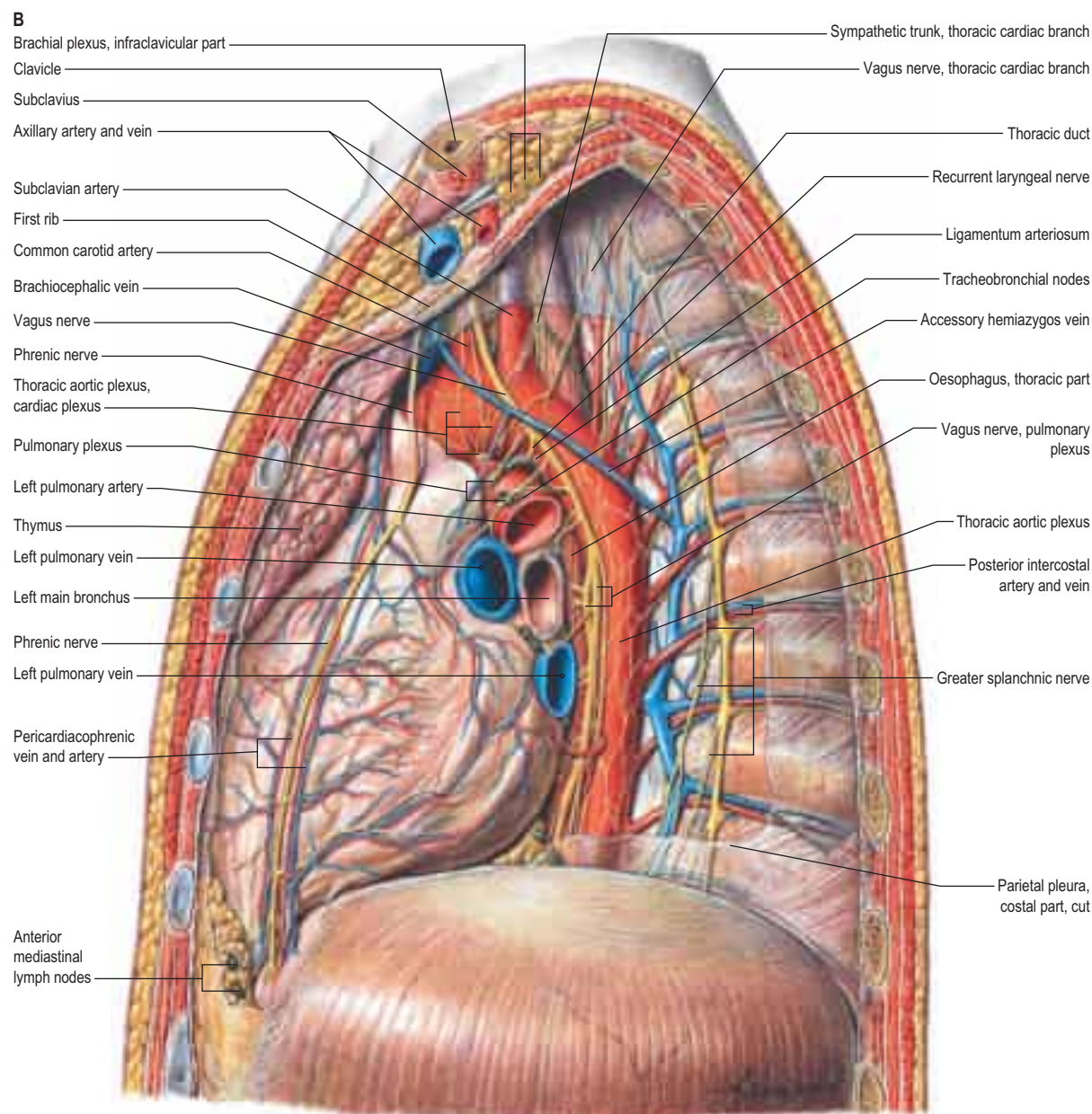


Fig. 56.7, cont'd The mediastinum. **B**, Left lateral aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

extent, the atrioventricular nodes (Ch. 57). Cardiac branches slow the heart rate and diminish the force of contraction. It has been claimed that the vagi indirectly influence ventricular muscle via their effect on the atrioventricular node, the postganglionic parasympathetic innervation of the ventricles being sparse. The smaller branches of the coronary arteries are innervated mainly via the vagus, whereas the larger arteries are mainly supplied by sympathetic fibres. Regional patterns of innervations may explain some particular forms of cardiomyopathies such as apical ballooning (Takotsubo cardiomyopathy). Pulmonary branches are motor to the circular smooth muscle fibres of the bronchi and bronchioles, and are therefore bronchoconstrictor; synaptic relays occur in the ganglia of the pulmonary plexuses.

The distribution of the vagi to the abdominal viscera is described in the relevant chapters.

Right vagus

The right vagus nerve descends posterior to the internal jugular vein in the neck (p. 466) and crosses the first part of the subclavian artery to enter the thorax; the ansa subclavia is lateral, and the phrenic nerve is further lateral. It then descends through the superior mediastinum, at first posterior to the right brachiocephalic vein, and then to the right of the trachea and posteromedial to the superior vena cava. Superiorly, the right pleura and lung are lateral; inferiorly, the azygos vein and its arching terminal portion separate the right vagus from the lung and pleura (see Fig. 56.7A). The nerve next passes posterior to the right main

bronchus and lies on the posterior aspect of the right pulmonary hilum. Here, it gives off posterior bronchial branches that unite with rami from the second to fifth or sixth thoracic sympathetic ganglia to form the right posterior pulmonary plexus. Two or three branches descend from the inferior part of this plexus on the posterior aspect of the oesophagus to join a left vagal branch and form the posterior oesophageal plexus. A vagal trunk containing fibres from both vagi leaves the plexus and runs inferiorly on the posterior surface of the oesophagus, entering the abdomen by passing through the diaphragmatic oesophageal aperture.

Left vagus

The left vagus enters the thorax posterior to the left brachiocephalic vein, between the left common carotid (anterior) and the subclavian arteries (posterior). It descends through the superior mediastinum and crosses the left side of the aortic arch, passing posterior to the left pulmonary hilum (see Fig. 56.7B). Superior to the aortic arch, it is crossed anterolaterally by the left phrenic nerve; on the arch, it is crossed by the left superior intercostal vein. Posterior to the left hilum, it divides into the posterior bronchial branches, which unite with rami of the second to fourth thoracic sympathetic ganglia to form the left posterior pulmonary plexus. Two or three branches descend anteriorly on the oesophagus and join with a ramus from the right posterior pulmonary plexus to form the anterior oesophageal plexus. A trunk containing fibres from both vagi descends anterior to the oesophagus and enters the abdomen

through the oesophageal diaphragmatic aperture. Numerous branches from the anterior vagal and left sympathetic trunks are distributed to the descending thoracic aorta. This pathway transmits the typical intense, excruciating pain of an intramural aortic haematoma from dissection or laceration.

THYMUS

The thymus and the bone marrow are the two primary lymphoid organs. The thymus is an encapsulated soft, bilobed organ; the two parts are joined in the midline by connective tissue that merges with the capsule of each lobe (Figs 56.8–56.9). Capsular tissue may have adhesions to the fibrous pericardium, which is thinner superiorly and may either easily tear or require a limited pericardotomy during thymectomy. The thymus is visible on CT and magnetic resonance imaging (MRI) just anterior to the ascending aorta and inferior to the left brachiocephalic vein. CT attenuation in younger individuals is homogeneous, greater than that of the chest wall and cardiac muscles below the age of 1 year, and similar to that of these muscles after the age of 1 year (Sklair-Levy et al 2000); the signal intensity with MRI on T₂-weighted images is similar to or greater than that of fat. The decline in CT attenuation with age is secondary to fatty infiltration.

Relations

The thymus is largest in the early part of life, particularly around puberty, and persists actively into old age despite considerable fibrofatty degeneration that sometimes hides the existence of persistent thymic tissue. The greater part of the thymus lies in the superior and anterior mediastina; the inferior aspect of the thymus reaches the level of the fourth costal cartilages. In some children, the thymus has been described

as encircling the left innominate vein (Han et al 1989). Superiorly, extensions into the neck are common, reflecting its bilateral embryonic origins from the third pharyngeal pouch (p. 619, see Fig. 36.18). Its superior poles join at, and extend above, the level of the suprasternal notch; the left usually extends higher and is seen first behind the strap muscles during the initial stages of transcervical thymectomy. It sometimes reaches the inferior poles of the thyroid gland or even higher, and is connected to the thyroid gland by the thyrothymic ligament. Its shape is largely moulded by adjacent structures. Inferiorly, the right lobe commonly lies between the right side of the ascending aorta and the right lung, anterior to the superior cava. Anteriorly, from superior to inferior, lie sternohyoid and sternothyroid, cervical fascia, the manubrium sterni, internal thoracic vessels and upper three costal cartilages. The pleurae lie laterally and the phrenic nerves are anterolateral and inferior; both structures may be injured during thymectomy. Posteriorly, the thymus is in contact with the vessels of the superior mediastinum (the left brachiocephalic vein and the inferior thyroid veins are usually partly embedded in the gland), the superior part of the thoracic trachea and the anterior cardiac surface (right atrium and ventricle). Thoracic surgeons performing thymectomy must be aware of the anatomical variation in which the upper poles may lie posterior to the left brachiocephalic vein.

Separated thymic tissue is often found scattered around the gland, and ectopic thymic rests are sometimes discovered in unusual mediastinal locations. Small accessory nodules may occur in the neck, representing separated portions, detached during embryological descent, and sometimes reaching more superiorly than the thyroid cartilage. Ectopic intrathyroidal thymi have been reported in children (Segni et al 2011).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

Thymic branches originating from the internal thoracic, inferior and sometimes superior thyroid arteries supply the thymus and the surrounding mediastinal connective tissue. As no definite hilum exists, the arterial branches either travel along the interlobar septa before entering the thymus at the junction of the cortex and medulla, or they reach the thymic tissue directly through the capsule.

Veins

Thymic veins drain to the left brachiocephalic, internal thoracic and inferior thyroid veins, and occasionally directly into the superior vena cava. One or more veins often emerge medially from each lobe of the thymus to form a common trunk opening into the left brachiocephalic vein and require careful ligation during thymectomy.

Lymphatic drainage

The thymus has no afferent lymphatics. Efferent lymphatics arise from the medulla and corticomedullary junction, drain through the extravascular spaces, accompany the supplying arteries and veins, and end in the brachiocephalic, tracheobronchial and parasternal nodes.

INNERVATION

The thymus is innervated from the sympathetic chains via the cervicothoracic (stellate) ganglia or ansa subclavia and from the vagi. Branches from the phrenic and descending cervical nerves (inferior roots of the ansa cervicalis) are distributed mainly to the capsule. The two lobes are innervated separately through their dorsal, lateral and medial aspects. During development and before its descent into the thorax, the thymus is innervated by the vagi in the neck. After its descent, the thymus receives a sympathetic innervation via fibres that travel along the vessels; postganglionic sympathetic terminations branch radially and form a plexus with the vagal fibres at the corticomedullary junction.

MICROSTRUCTURE

The thymus is responsible for the provision of thymus-processed (T) lymphocytes to the entire body, and provides a unique microenvironment in which thymocytes (T-cell precursors) undergo development, differentiation and clonal expansion to deliver the exquisitely specific

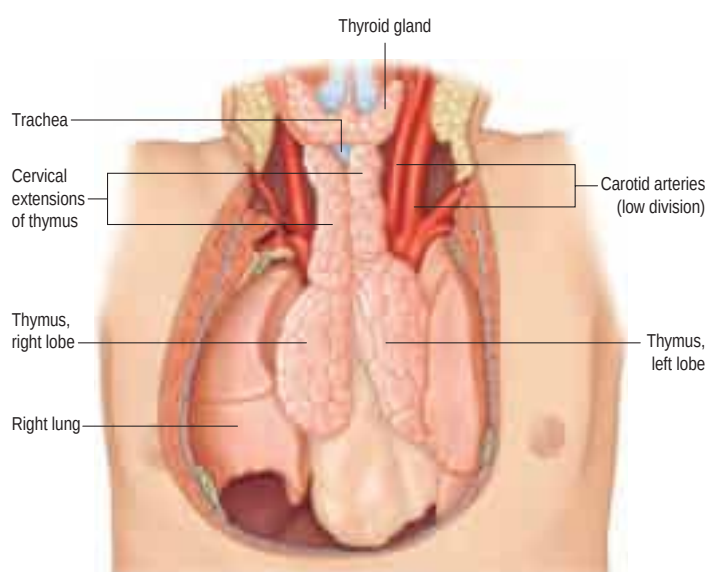


Fig. 56.8 The neonatal thymus.



Fig. 56.9 A thymus from a 9-year-old female (left) and one from an 80-year-old male (right). Note the fibrofatty infiltration of the older thymus. (Courtesy of Professor M Kendall.)



The roles of the nervous system and other neuroendocrine elements in the overall biology of the thymus are little understood. Many of the autonomic nerves are doubtless vasomotor, but other terminal branches (at least in rodents) ramify among the cells of the thymus, particularly the medulla, suggesting that they may have other roles. Innervation is complete by the onset of thymic function. The medulla contains a number of different types of non-lymphoid cells, including cells positive for vasoactive intestinal polypeptide and acetylcholinesterase; large non-myoid cells; and cells containing oxytocin, vasopressin and neurophysin, of possible neural crest origin.

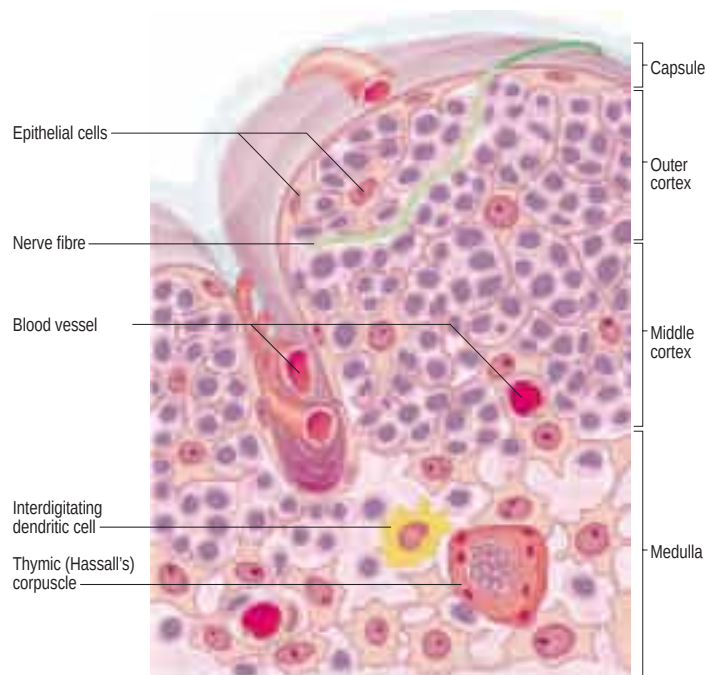


Fig. 56.10 Thymic structure and cellular organization, showing an interlobular septum, the cortical circulation and thymocytes within an epithelial framework.

T-cell response, simultaneously acquiring immune tolerance to the body's own components. These steps involve intimate interactions between thymocytes, mainly epithelial and antigen-presenting cells and chemical factors in the thymic environment. The thymus is also part of the neuroimmunological and neuroendocrine axes of the body, influenced by and influencing the products of these axes. Its activity therefore varies throughout life under the influence of different physiological states, disease conditions and chemical insults, such as hormones, drugs and pollutants. The thymus appears to be most active between the ages of 3 and 6 months, when the highest number of total thymocytes are present (Weerkamp et al 2005).

General architecture

It is useful to consider the embryological origins of the thymus in order to understand its cellular organization. It is derived from a number of sources, including epithelial derivatives of the pharyngeal pouches, mesenchyme, haemolymphoid cells and vascular tissue. In section, the thymus can be seen to consist of an outer cortex of densely packed thymocytes (T-cell precursors) and an inner medulla with fewer lymphoid cells (Fig. 56.10). Both thymic lobes have a loose, fibrous connective tissue capsule, from which septa penetrate to the junction of the cortex and medulla, and partially separate the irregular lobules, which are each 0.5–2.0 mm in diameter (see Fig. 56.12). The connective tissue septa form a route of entry and exit for blood vessels and nerves, and carry efferent lymphatics. Most migrant cells enter or leave the thymus by this route. In each lobule, the cortex is composed of a superficial subcapsular cortex (a narrow band of cells immediately beneath the capsule) and a more extensive main cortex. The central medulla of both thymic lobes is continuous from one lobule to the next.

Epithelial framework

Unlike other lymphoid structures, in which the supportive framework is chiefly collagenous reticular tissue, the thymus contains a network of interconnected epithelial cells (see Fig. 56.10) that create an appropriate microenvironment, by cell-to-cell contact and the release of paracrine factors, in which thymic lymphocytes (T cells) develop and mature. Although differing in morphology, all the epithelial cells of the thymus share a common origin from pharyngeal endoderm. They vary in size and shape, according to their positions within the thymus. Typically, they have pale, oval nuclei, a rather eosinophilic cytoplasm and intercellular desmosomal attachments. Intermediate filament bundles of cytokeratin lie within their cytoplasm. The subcapsular cells form a continuous external lining to the thymus, following its lobulated profile, ensheathing the penetrating vessels, and contributing to the blood–thymus barrier. Other cortical epithelial cells form a loose mesh

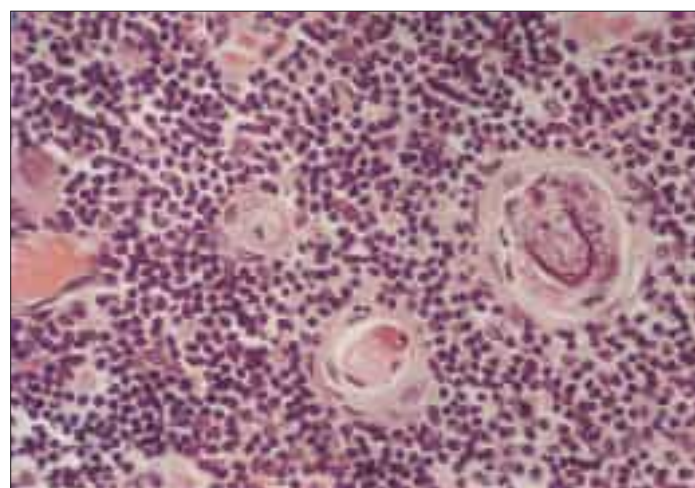


Fig. 56.11 The medulla of a neonatal thymus, showing three concentric Hassall's corpuscles of varying degrees of maturity, surrounded by closely packed lymphocytes and a few epithelial cells with larger nuclei.

of long cytoplasmic processes, whereas medullary epithelial cells tend to form more solid cords and Hassall's corpuscles. Lymphocytes lie within the interstices of the mesh or between the cords. Large epithelial 'nurse' cells may be associated with around 50 or more thymocytes.

Hassall's corpuscles are whorls of flattened, concentrically layered cells, 30–100 µm in diameter, that characterize the thymic medulla (see Fig. 56.10; Fig. 56.11). Originating prenatally, and increasing in number throughout life, they perhaps remove dying thymocytes because their centres are eosinophilic and partly keratinized, and often contain cellular debris. Corpuscles with a similar appearance have been described in the palatine tonsil. (For further reading, see Raica et al (2006), Asghar et al (2012).)

Other non-lymphocytic thymic cells

The thymus also contains cells of the myeloid, fibroblastic and myoid lineages. Myeloid lineage cells include monocytes at the corticomedullary junction, mature macrophages throughout but particularly in the cortex, and interdigitating dendritic (antigen-presenting) cells at the corticomedullary junction and in the medulla. Some dendritic cells are of lymphoid rather than myeloid origin (see Fig. 4.12). Fibroblasts are found in the capsule, perivascular spaces and medulla but are infrequent in the cortex, except in the involute thymus. Myoid cells, which are relatively rare, are situated mainly in the medulla and at the corticomedullary junction. They are large, rounded cells possessing a central nucleus surrounded by irregularly arranged bundles of myofilaments with unknown function, although it has been suggested that their contractions might aid the movement of lymphoid cells across or out of the thymus.

Thymocytes

The cortex is densely packed with small thymocytes (thymic lymphocytes, presumptive T cells; see Ch. 4). They occupy the interstices of the epithelial reticulum, which they largely obscure in histological sections, and form 90% of the total weight of the neonatal thymus. A distinct subcapsular zone houses the thymic stem cells and lymphoblasts undergoing mitotic division. The first stem cells to enter the thymus in the embryo come from the yolk sac and liver during their haemopoietic phases. During later developmental periods, it is probable that all thymic lymphocytes originate in the bone marrow before passing in the blood stream to the thymus.

Thymocytes undergo mitosis in all cortical zones as the differentiating T cells mature, gradually moving deeper into the cortex (Fig. 56.12). The processes of thymocyte development and maturation to generate T cells depend on the microenvironment provided by epithelial cells, dendritic cells, macrophages and fibroblasts. T cells that either fail to recognize self-MHC (histocompatibility) molecules or that recognize self-antigens die by apoptosis, in order to achieve functional immune reactivity and maintain self-tolerance, respectively. Over 95% of cortical thymocytes die within the thymus; the surviving T cells migrate through the walls of venules and efferent lymphatics to enter the circulation and populate secondary lymphoid tissues (for further details of thymocyte development, see p. 76).

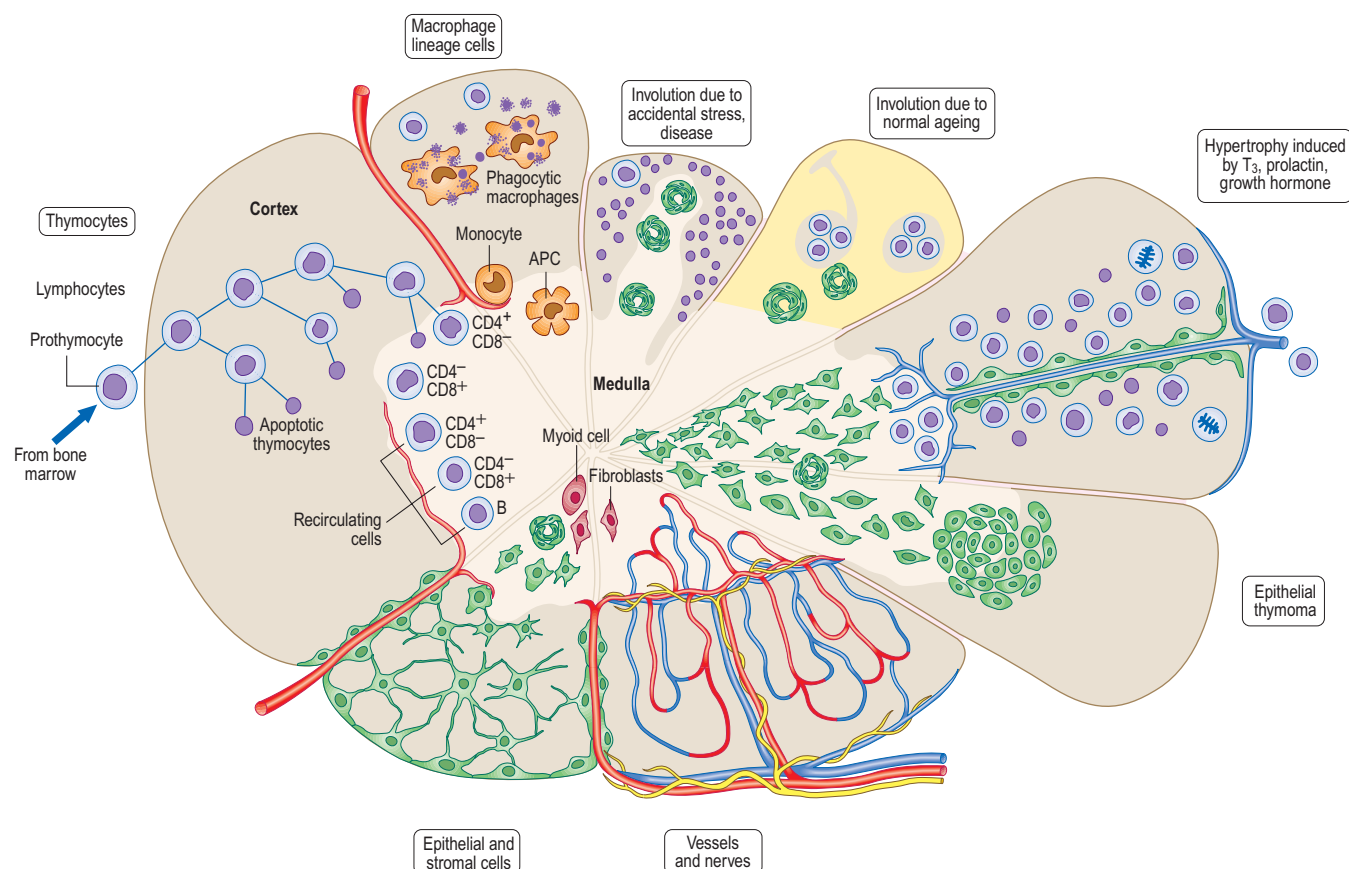


Fig. 56.12 The microscopic organization of the thymus at various stages of life and under different conditions. Abbreviations: APC, antigen-presenting cell; T_3 , tri-iodothyronine hormone.

Microcirculation

Cortex

The pattern of blood flow differs in the cortex and medulla. Major blood vessels enter the gland at the corticomedullary junction and pass within each lobe, giving off small capillaries to the cortex and larger vessels to the medulla. Most cortical capillaries loop around at different depths in the cortex and join venules at the corticomedullary junction; some continue through the cortex and join larger venules running in the capsule that leave the thymus. Sheaths of thymic epithelial cells of the blood–thymus barrier lie between the narrow perivascular space of the smaller cortical capillaries and cortical thymocytes (Henry et al 1992). This perivascular space sometimes contains pericytes and, rarely, neurones.

Medulla

Medullary blood vessels are not as well protected by epithelial cells, and those of the corticomedullary junction are only partially ensheathed, usually on their cortical aspect. Medullary vessels are very variable in size; some have short lengths of cuboidal endothelium similar to those in lymph nodes and mucosa-associated lymphoid tissue (MALT; p. 75).

DEVELOPMENT

The embryology and prenatal development of the thymus are described on page 619.

Thymic changes during postnatal life

At birth, the thymus is most often bilobar and is 4–6 cm long, 2.5–5 cm wide and 1 cm thick. The bulkiest part of the gland at this age is immediately above the base of the heart where it covers the phrenic nerves laterally; the protection of these nerves is mandatory during paediatric cardiac surgery procedures. The left lobe is longer and thicker than the right lobe in full-term neonates (Varga et al 2011); during childhood, the thymus narrows and lengthens, and the cervical portion becomes less noticeable. Imaging studies of the thorax in normal children reveal that the right lobe of the thymus typically measures 9 mm in thickness and the left 11 mm. After the age of 20 years, it decreases to 5–6 mm in thickness (see Fig. 56.12).

The thymus is largest relative to body mass at birth, weighing 10–15 g. It is typically larger in boys than girls at 6 months of age (Garly et al 2008). Studies of thymic mass after sudden death have recorded a wide variation at all ages, but the general pattern is that, after increasing in the first year of life, the mean mass remains fairly constant at 20 g until the sixth decade, when a reduction occurs. At birth, individual adipocytes are present in connective tissue septa, and increased numbers are found within the cortex in the second and third decades. Fatty infiltration is usually complete by the fourth decade, when only the medulla and small patches of associated cortex are spared. This process is independent of body mass index (BMI).

In children, the gland is more pyramidal in shape and firmer than in later life, when the amount of lymphoid tissue is greatly reduced. In the fresh state, it is deep red, reflecting its hypervascularity (see Fig. 56.9). With age, it becomes thinner, greyer and increasingly infiltrated by yellow adipose tissue. Each of the two lobes is partially divided by the ingrowth of shallow septa, so that, superficially, the gland appears lobulated. As fatty atrophy proceeds, this lobulation becomes more distinct. The older thymus may be distinguished from the surrounding mediastinal fat only by the presence of its capsule. However, even greatly atrophied glands usually contain perivascular areas of persistent lymphoid tissue. Thymocyte production and differentiation persist throughout life, and thymic T cells continue to populate peripheral lymphoid tissue, blood and lymph.

Congenital anomalies of the thymus

Undescended thymus, accessory thymic bodies and rare cysts of the third branchial pouch are usually of no clinical significance in isolation. Patients with thymic agenesis, aplasia and hypoplasia, as in severe combined immune deficiency disease (SCID), have reduced lymphocyte numbers, and early death from infection is common. Most cases are familial, with autosomal recessive genes. In young children, a large normal thymus may press on the trachea, causing attacks of ventilatory stridor. This condition must be differentiated from other congenital conditions such as vascular rings and slings that require surgical correction.

THYMECTOMY

The thymus is essential for the normal development of lymphoid tissues during neonatal and early postnatal life. Thymectomy during



this period results in hypoplasia of the peripheral lymphoid organs, wasting and an inability to mount an effective immune response; it is ultimately fatal. By the onset of puberty, when the main lymphoid tissues are fully developed, thymectomy is less debilitating, but effective responses to novel antigens are reduced.

Thymectomy is commonly performed for myasthenia gravis, usually after referral from a neurologist. The most beneficial effects of thymectomy are seen in myasthenic females who present with systemic muscular weakness; controversial operative groups include the elderly and those with isolated ocular symptoms.

Thymoma and myasthenia gravis



Available with the *Gray's Anatomy* e-book

OESOPHAGUS

The oesophagus is a muscular tube, 25 cm long in the adult, that connects the pharynx to the stomach (see Figs 56.4, 56.6; Figs 56.13–56.14). Beginning in the neck, level with the lower border of the cricoid cartilage and the sixth cervical vertebra, it descends largely anterior to the vertebral column through the superior and posterior mediastina, passes through the diaphragm, level with the tenth thoracic vertebra,

and ends at the gastric cardiac orifice, level with the eleventh thoracic vertebra.

Generally vertical in its course, the oesophagus normally displays two shallow curves. Starting in the median plane, it inclines to the left as far as the root of the neck, gradually returns to the median plane near the fifth thoracic vertebra, then deviates leftwards again at the seventh thoracic vertebra before piercing the diaphragm. The oesophagus also deviates sagittally as it follows the cervicothoracic vertebral lordosis. Further inferiorly, it is pushed slightly to the right by the aorta before resuming its leftward course towards the oesophageal hiatus. It is the second narrowest part of the alimentary tract (the narrowest is the vermiform appendix) and is constricted at its start (15 cm from the incisor teeth, as measured during oesophagoscopy); where it is crossed anteriorly by the aortic arch (23 cm from the incisor teeth, as measured during oesophagoscopy) and by the left principal bronchus (28 cm from the incisor teeth, as measured during oesophagoscopy); and as it passes through the diaphragm (40 cm from the incisor teeth, as measured during oesophagoscopy). In children, the length of the oesophagus correlates best with height (Yang et al 2005).

CERVICAL OESOPHAGUS

The cervical portion of the oesophagus is posterior to the trachea and attached to it by loose connective tissue. The recurrent laryngeal

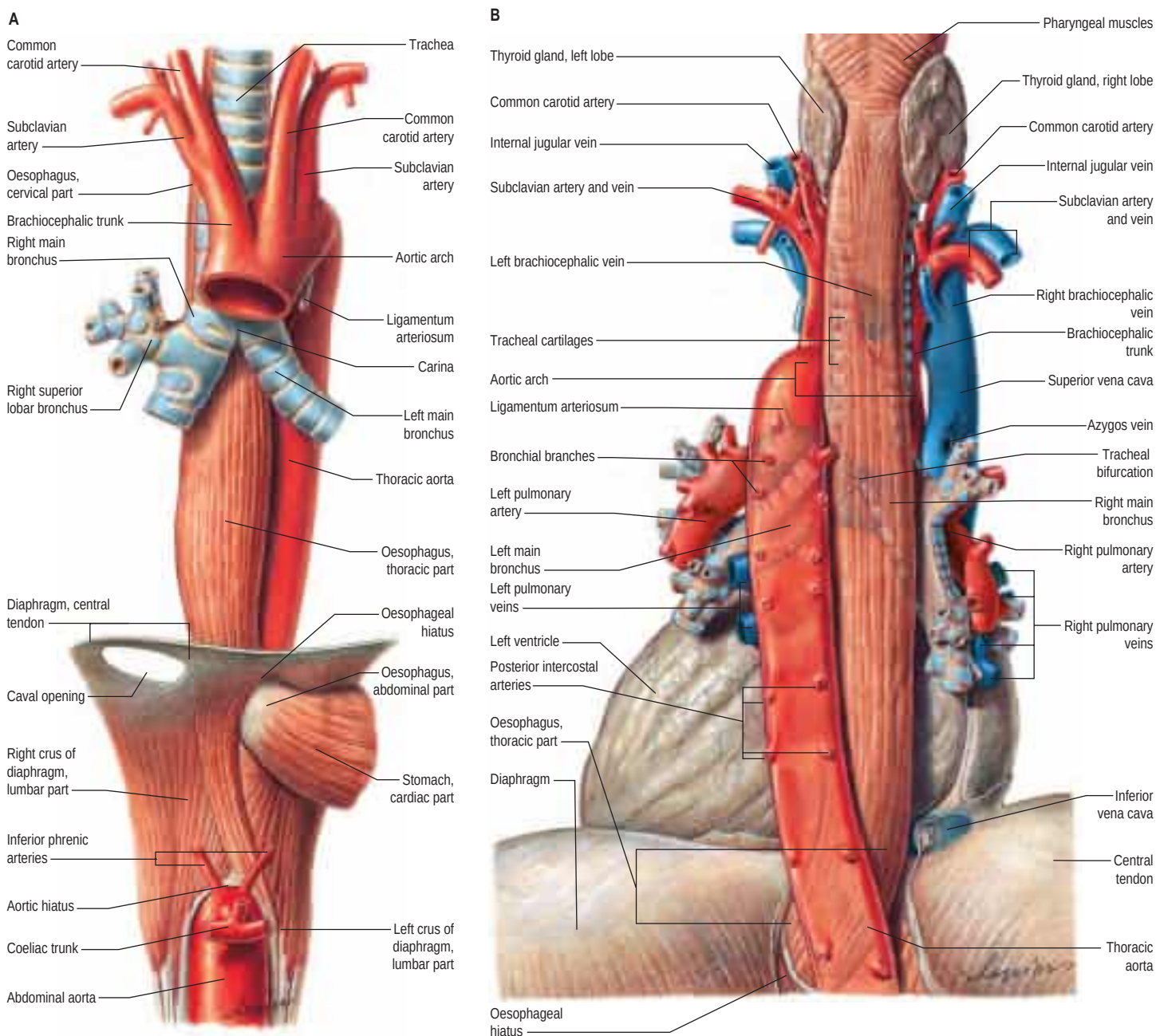


Fig. 56.13 The oesophagus. **A**, Anterior view. **B**, Posterior view. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Thymectomy is performed through a transcervical approach or a sternotomy, the choice of approach depending on the size and disposition of the diseased gland. The cervical approach involves a lower transverse incision that includes the sternal heads of both sternocleidomastoids, permitting an improved access to the substernal plane. Sternothyroid and sternohyoid are separated and the superior poles of the thymus are identified immediately underneath these muscles. A safe pre-thymic plane is created by blunt dissection just posterior to the sternum, facilitating subsequent mobilization of the gland. The mediastinal great vessels and the thymic vessels must be carefully identified. The thymic veins are selectively ligated and divided, especially those draining directly into the superior vena cava or the left brachiocephalic vein. The smaller arterial branches from the internal thoracic arteries are usually dealt with using electrocautery, care being taken to avoid injury to the phrenic nerves (patients are generally not given a paralytic agent during anaesthesia, in order for the phrenic nerves to be identified intraoperatively). The thymus generally remains completely encapsulated and is separated without difficulty from the pericardium (occasionally, adhesions necessitate a limited pericardiectomy). Further inferiorly, a gentle push on the great vessels allows direct visualization into the aortopulmonary window for complete removal of the thymus in this vicinity.

In most cases, the thymus gland is removed completely with both upper and lower poles intact. Careful inspection of the remaining tissue in the mediastinum is performed to identify any possible anatomical anomalies that might result in retained thymic tissue after the operation. The most common anomaly is the unanticipated location of the upper poles of the thymus behind the left brachiocephalic vein. The aortopulmonary window is also a common location for thymic tissue and this area is sometimes difficult to expose with a transcervical approach. Any suspicious foci of fat in the mediastinum are removed, and, if necessary, sent for frozen section to ascertain whether or not they contain thymic tissue.

Thymic tumours may compress the trachea, oesophagus and large veins in the neck, causing hoarseness, cough, dysphagia and venous engorgement of the head and neck. Thymomas tend to have a heterogeneous signal intensity on MRI, often developing in one lobe of the thymus without affecting the other. Many affected patients also have myasthenia gravis and other autoimmune conditions. Myasthenia gravis, a chronic autoimmune disease of adults, presents as a diminution in power of repetitive contraction in certain voluntary muscles. Although there may be more than one condition with these signs, myasthenia gravis is essentially an autoimmune disease in which acetylcholine receptor proteins of neuromuscular junctions are attacked by autoantibodies. Muscles commonly involved are levator palpebrae superioris, leading to ptosis, and the extraocular muscles, leading to diplopia. Other muscles in the face, jaws, neck and limbs may be involved, and in severe cases, the ventilatory muscles are compromised. Occasional Caucasian individuals with myasthenia gravis have a thymoma and some have medullary follicular hyperplasia; the latter are predominantly females younger than 40 years of age with strong expression of HLA-B8-DR3, in whom thymectomy often results in symptomatic improvement. In the absence of a thymoma, the onset of myasthenia gravis occurs after 40 years of age in patients with an HLA-B7-DR2 phenotype, except for a group in whom weaknesses are restricted to eye and eyelid movements.

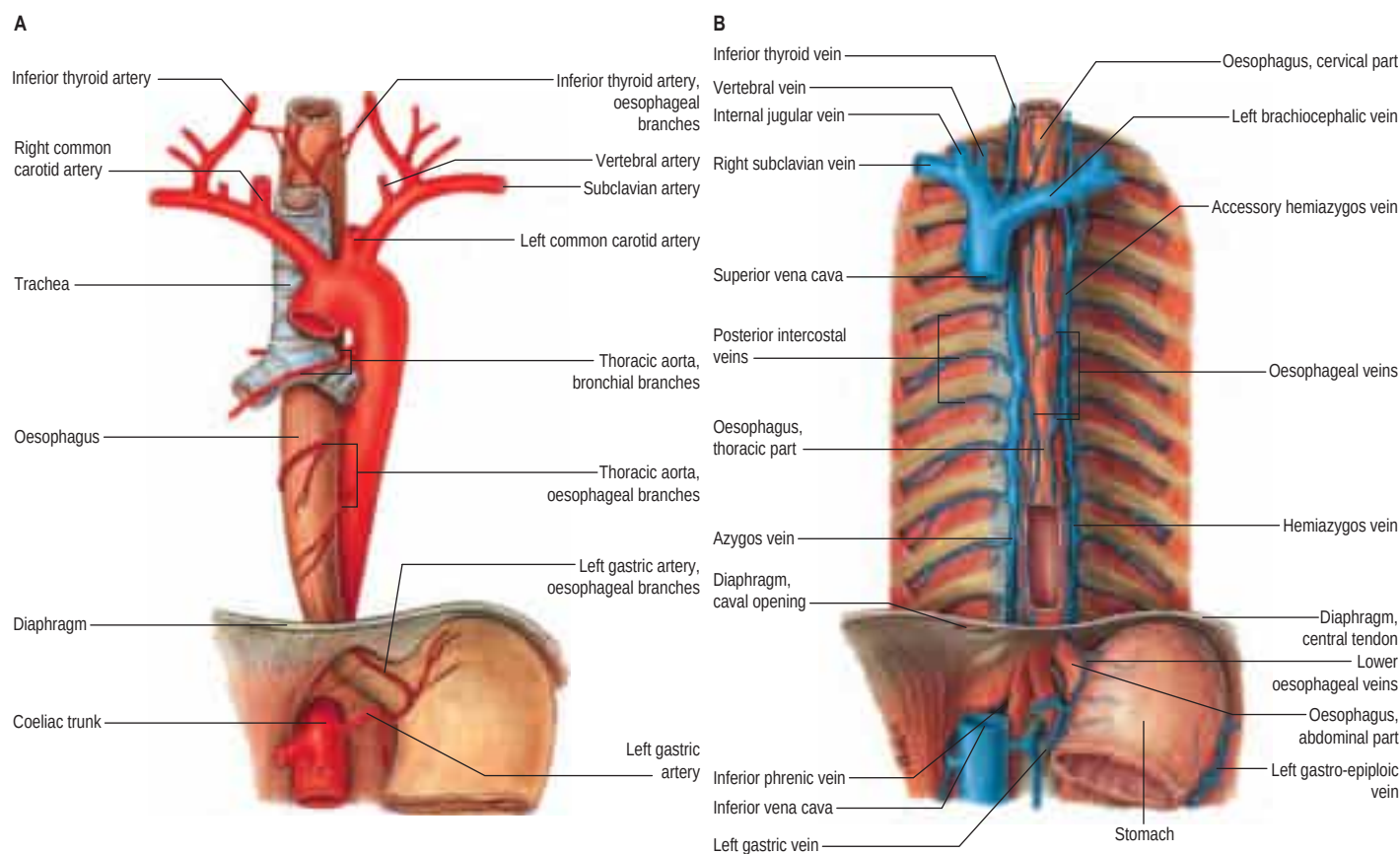


Fig. 56.14 The oesophagus, anterior view. **A**, Arterial blood supply. **B**, Venous drainage. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

nerves ascend on each side in or near the tracheo-oesophageal groove. Posteriorly are the vertebral column, longus colli and the pre-vertebral layer of deep cervical fascia. Laterally, on each side, are the common carotid arteries and the posterior part of the thyroid gland. In the lower neck, where the oesophagus deviates to the left, it is closer to the left carotid sheath and thyroid gland than it is on the right. The thoracic duct ascends for a short distance along its left side (see Figs 56.5, 56.7B).

THORACIC OESOPHAGUS

The thoracic oesophagus (see Figs 56.13–56.14) is situated a little to the left in the superior mediastinum between the trachea and the vertebral column. It passes behind and to the right of the aortic arch to descend in the posterior mediastinum along the right side of the descending thoracic aorta. Below, as it inclines left, it crosses anterior to the aorta and enters the abdomen through the diaphragm at the level of the tenth thoracic vertebra. From above downwards, the trachea, right pulmonary artery, left principal bronchus, pericardium (separating it from the left atrium) and the diaphragm are anterior. The vertebral column, longus colli, right posterior intercostal arteries, thoracic duct, azygos vein and the terminal parts of the hemiazygos and accessory hemiazygos veins, and, near the diaphragm, the aorta are posterior. A long recess of the right pleural sac lies between the oesophagus anteriorly and the azygos vein and vertebral column posteriorly in the posterior mediastinum (the azygo-oesophageal recess).

In the superior mediastinum, the terminal part of the aortic arch, left subclavian artery, thoracic duct, left pleura and the recurrent laryngeal nerve are left lateral relations. In the posterior mediastinum, the oesophagus is related to the descending thoracic aorta and left pleura. The right pleura and the azygos vein, as it arches forwards above the right main bronchus to join the superior vena cava, are right lateral relations. Below the pulmonary roots, the vagus nerves descend in contact with the oesophagus, the right mainly behind and the left in front; the vagi subsequently unite to form a plexus around the oesophagus. Low in the posterior mediastinum, the thoracic duct is posterior and to the right of the oesophagus; at higher levels, the duct is posterior, crossing to the left of the oesophagus at about the level of the fifth thoracic vertebra, then ascending on the left.

On the right of the oesophagus, just above the diaphragm, a small, serous infracardiac bursa may occur, representing the detached apex of the right pneumatoenteric recess.

The close relationship between the oesophagus, the heart and the aorta constitutes the anatomical basis of transoesophageal cardiovascular ultrasonography. Aortic dissection may be seen in the descending thoracic aorta in cases of equivocal CT and MRI.

ABDOMINAL OESOPHAGUS

The abdominal oesophagus is described in Chapter 64.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The cervical oesophagus is supplied by the inferior thyroid artery (see Fig. 56.14, p. 458). The thoracic oesophagus is supplied by bronchial and oesophageal branches of the thoracic aorta (see Fig. 56.14A). Four or five oesophageal branches arise from the anterior surface of the aorta and descend obliquely to the oesophagus, where they form a vascular chain that anastomoses above with the oesophageal branches of the inferior thyroid arteries, and below with ascending branches from the left phrenic and left gastric arteries. Extensive devascularization of the oesophagus may lead to acute oesophageal necrosis as in aortic dissection (van der Wal-Visscher et al 2011).

Veins

Blood from the oesophagus drains into a submucous plexus and thence into a peri-oesophageal venous plexus, from which the oesophageal veins arise. Those from the thoracic oesophagus drain predominantly into the azygos veins and, to a lesser extent, the hemiazygos, intercostal and bronchial veins (see Fig. 56.14B). Those from the cervical oesophagus drain into the inferior thyroid vein. The left gastric vein meets the lower oesophageal veins at the oesophageal opening in the lesser curvature, one of the sites of porto-caval venous anastomoses where variceal transformation can occur in portal hypertension (see below).

Oesophageal varices

Cirrhosis or fibrosis of the liver affects the vascular tree within the liver, resulting in a decrease of hepatic vascular compliance, and a likely increased vascular tone reflecting a reduction in endothelial vasodilators such as nitric oxide. The portal resistance increases, leading to the formation of a collateral circulation, and there is a concomitant increase in the systemic and splanchnic blood flow. Porto-systemic shunting of blood occurs between the short gastric coronary veins and the oesophageal veins, largely as the result of the dilation of pre-existing embryonic channels.

Lymphatic drainage

The oesophagus has an extensive, longitudinally continuous submucosal lymphatic system, which presumably explains remote para-oesophageal lymphadenopathy in oesophageal carcinoma. Efferent vessels from the cervical oesophagus drain to the deep cervical nodes either directly or through the paratracheal nodes. Vessels from the thoracic oesophagus drain to the posterior mediastinal nodes and those from the abdominal oesophagus drain to the left gastric lymph nodes. Some may pass directly to the thoracic duct.

INNERVATION

The upper oesophagus is supplied by the branches of the recurrent laryngeal nerve and by postganglionic sympathetic fibres that reach it by travelling along the inferior thyroid arteries. The lower oesophagus is supplied by the oesophageal plexus, a wide-meshed autonomic network that surrounds the oesophagus below the level of the lung roots, and contains a mixture of parasympathetic and sympathetic fibres (see Fig. 56.6).

Motor fibres to the striated and smooth muscle of the oesophageal wall travel in the vagus. Axons derived from neuronal cell bodies in the nucleus ambiguus travel via the recurrent laryngeal nerve and supply cricopharyngeus and the striated muscle of the upper third of the oesophagus. Axons with cell bodies in the dorsal nucleus of the vagus pass through the oesophageal plexus and supply the smooth muscle that makes up the lower two-thirds of the oesophagus, after local relay in the oesophageal wall. Other branches are given off by the vagus as it travels through the mediastinum, and pass directly to the oesophagus. The vagus also carries secretomotor fibres to mucous glands in the oesophageal mucosa, and visceral afferent (sensory) fibres to cell bodies in its inferior ganglion.

Vasomotor sympathetic fibres destined for the oesophagus arise from the upper 4–6 thoracic spinal cord segments. Those from the upper ganglia pass to the middle and inferior cervical ganglia, where they synapse on postganglionic neurones that give rise to axons that innervate the vessels of the cervical and upper thoracic oesophagus. Those from the lower ganglia pass either directly to the oesophageal plexus or to the coeliac ganglion (via the greater splanchnic nerve), where they synapse; postganglionic axons innervate the distal oesophagus. Afferent visceral pain fibres travel via the sympathetic fibres to the first four segments of the thoracic spinal cord. Because these segments also receive cardiac afferents, it is sometimes difficult to distinguish between oesophageal and cardiac pain.

MICROSTRUCTURE

The tissues forming the thoracic oesophageal wall, from the lumen outwards, are the mucosa (consisting of epithelium, lamina propria and muscularis mucosae), submucosa, muscularis externa and adventitia (Fig. 56.15).

Mucosa

The mucosa is thick. At the gastro-oesophageal junction, a jagged boundary line separates the greyish-pink smooth oesophageal mucosa from the reddish-pink gastric mucosa, the latter covered by minute bulges and depressions. Throughout its length, the oesophageal lumen is marked by deep longitudinal grooves and ridges, which disappear when the lumen is distended, otherwise obliterating it.

Epithelium

The epithelium is non-keratinized, stratified squamous, and is continuous with that of the oropharynx. In humans, this protective layer is quite thick (300–500 µm) and is not affected by oesophageal disten-

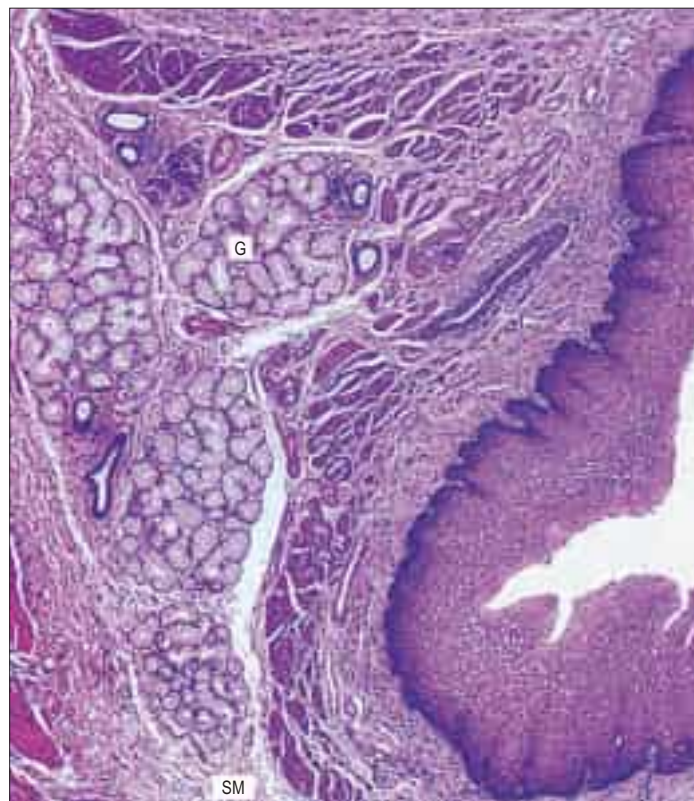


Fig. 56.15 The wall of the oesophagus. A stratified squamous, non-keratinizing epithelium lines the lumen (right). Submucosal glands (G) in the submucosa (SM) secrete mucus that lubricates the passage of food. (With permission from Dr JB Kerr, Monash University, from Kerr JB 1999 Atlas of Functional Histology. London: Mosby.)

sion. The boundary between the oesophageal epithelium and its lamina propria is distinct but markedly uneven due to tall connective tissue papillae invaginating the epithelial base, assisting in the anchorage of the epithelium to underlying tissues (see Fig. 56.15). These papillae are permanent structures, also unaffected by oesophageal distension, and they are rich in blood vessels and nerve fibres. At the base of the epithelium, there is a basal lamina, to which epithelial cells are attached by hemidesmosomes, as occurs in the oral mucosa.

Oesophageal epithelium is similar to other stratified squamous epithelia. It can be divided into a basal, proliferative layer, a parabasal layer of cells undergoing terminal differentiation and a flattened layer of superficial cells or squames that retain their nuclei. The most superficial strata of cells contain a few keratohyalin granules, in addition to keratin filaments. The epithelial cell population is constantly renewed by mitosis in the cuboidal basal and deepest parabasal cells; as they migrate towards the lumen, they become progressively polygonal and then more flattened, eventually desquamated at the epithelial surface. This sequence of events normally takes 2–3 weeks, and is markedly slower than in the stomach and intestine.

The epithelium is an effective protection against mechanical injury during swallowing because of its thickness and the presence of mucus at its surface. However, protection is limited by repeated exposure to the strongly acidic, protease-rich secretions of the stomach, as occurs abnormally during reflux. Failure of the lower oesophageal sphincter allowing reflux results in ulceration and fibrosis, accompanied by dysphagia/odynophagia, with possible epithelial metaplasia to a gastric-like mucosa (Barrett's mucosa), or to more overt neoplastic changes.

Langerhans cells Langerhans cells are present in the oesophageal epithelium. They are immature dendritic cells and resemble those found in the epidermis. They perform similar antigen-processing and antigen-presenting roles, which are important in immunostimulation of naïve T cells and mucosal defence.

Lamina propria

The oesophageal lamina propria contains scattered groups of mucosa-associated lymphoid tissue (MALT), especially prominent near the gastro-oesophageal junction. Small tubular mucous glands occur in this region and also at the oesophago-pharyngeal junction.

Varices in the distal oesophagus are easily visible at endoscopy because they are situated superficially in the lamina propria. The blood from the superficial veins drains into a superficial venous plexus, then into a deeper intrinsic venous plexus and finally into the perioesophageal veins via perforating veins. Bidirectional flow is normally possible in this region, a phenomenon that permits pressure changes during breathing and Valsalva manoeuvres. However, in portal hypertension, the valves within the perforating vessels become incompetent and blood flow is retrograde, causing dilation of the deep intrinsic veins. The greater pressure within this region predisposes the varices to brisk and life-threatening bleeding. Treatment is directed towards controlling the formation of a collateral circulation and the obliteration of varices that are susceptible to bleeding, and is achieved by paravariceal endoscopic injection of a sclerosant, which causes obliteration of the varices as a result of thrombus formation. Fibrosis is also induced within the mucosa, which reduces the formation of new collateral vessels. Alternatively, rubber bands may be applied in an attempt to ligate the varices. Bleeding from varices is associated with a mortality rate of 25%, reflecting the problems of rebleeding and underlying comorbidity.

Muscularis mucosae

The muscularis mucosae is composed mainly of longitudinal smooth muscle and forms a thin sheet near the epithelium, the contours of which it follows closely. At the pharyngeal end of the oesophagus, it may be absent or represented only by sparse, scattered bundles; below this, it becomes progressively thicker. The longitudinal orientation of its cells changes to a more plexiform arrangement near the gastro-oesophageal junction. The ducts of the oesophageal glands pierce the muscularis mucosae.

Submucosa

The submucosa loosely connects the mucosa and the muscularis externa, penetrating the longitudinal ridges of the oesophageal lumen. It contains larger blood vessels, nerves and mucous glands. Its elastic fibres aid closure of the oesophageal lumen after peristaltic dilation.

Oesophageal glands

Groups of small tubule-acinar oesophageal glands mainly lie in the submucosa, each sending a single long duct through the intervening layers into the lumen. They are composed mostly of mucous cells, with some serous cells that secrete lysozyme. Close to the pharynx and stomach, the glands are simpler in form and restricted to the lamina propria of the mucosa (not submucosa). The mucosal mucous glands of the abdominal oesophagus closely resemble the gastric cardiac glands, and are therefore named oesophageal cardiac glands.

Muscularis externa

The muscularis externa is up to 300 µm thick, and consists of the outer longitudinal and inner circular layers typical of the intestine. The thicker longitudinal layer exists as a continuous coat of fibres that extends almost the entire length of the oesophagus, except posterosuperiorly, 3–4 cm inferior to the cricoid cartilage, where it diverges as two fascicles that ascend obliquely and anteriorly, passing deep to the lower border of the inferior constrictor, and ending in a tendon that attaches superiorly to the posterior ridge of the cricoid lamina. The V-shaped space between these fascicles is filled by the circular oesophageal muscle fibres, thinly covered inferiorly by decussating longitudinal fibres and superiorly by the inferior constrictor, predisposing to the oesophageal diverticulum (Zencker).

Accessory slips of smooth muscle sometimes pass between the oesophagus and left pleura or the root of the left principal bronchus, trachea, pericardium or aorta. Some consider these slips to fix the oesophagus to these structures, but they actually equilibrate the normal physiological changes in position, shape and dimensions of the various organs between which they extend. Superiorly, the circular fibres blend posteriorly with the inferior pharyngeal constrictor, and anteriorly, the uppermost fibres attach to the lateral margins of the tendon of the two longitudinal fasciculi of the oesophagus. Inferiorly, the circular muscle is continuous with the oblique layer of muscle fibres in the stomach wall.

In the upper third of the oesophagus, the muscularis externa is formed by skeletal muscle; in the middle third, smooth muscle fascicles intermingle with striated muscle; and this increases distally such that the lower third contains only smooth muscle. This is also reflected by the innervation pattern of the oesophagus (see above).

OESOPHAGOSCOPY AND TRANSOESOPHAGEAL ULTRASOUND



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BARIUM STUDIES



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DEVELOPMENT

The development of the oesophagus is described on [page 1048](#).

Tracheo-oesophageal fistula

Tracheo-oesophageal fistula is described on [page 930](#).

OESOPHAGEAL SPHINCTERIC MECHANISMS

Radiological studies show that swallowed food stops momentarily in the gastric end of the oesophagus, before entering the stomach (see [Fig. 56.18](#)), suggesting the presence of a sphincter at this point. In the past, there was much controversy surrounding the reason for this behaviour because only slight thickening of the muscle coat has been found in humans. There is now ample physiological and clinical evidence that closure depends on two major mechanisms. The more important of these is the lower oesophageal sphincter, a specialized zone of circular smooth muscle surrounding the oesophagus at its transit through the diaphragm and for much of its short abdominal course. This region of the oesophagus is maintained under tonic contraction, except during swallowing, when it relaxes briefly to admit ingesta to the stomach, and during vomiting. Intramural plexuses of the enteric nervous system and the neural release of nitric oxide contribute to its relaxation. The second mechanism is a functional external sphincter provided by the right crus of the diaphragm, which encircles the oesophagus as it passes into the abdomen and is attached to it by the phreno-oesophageal ligament. Fluoroscopy, electromyography and manometry have shown that its muscular fibres contract around the oesophagus during inspiration and when intra-abdominal pressure is increased, thus helping to prevent gastro-oesophageal reflux, even when the lower oesophageal sphincter is inhibited experimentally with atropine. The relative importance of these two agents in the prevention of oesophageal reflux is still a matter for debate. Clinically, there is a good correlation between this condition and lower oesophageal sphincter dysfunction in some cases, whereas in others, failure of the diaphragmatic component, as seen in hiatus hernia, appears to be a major factor. The anatomical configuration of the gastro-oesophageal orifice may also play some part in these processes.

Swallowing

The oesophageal phase of swallowing is described on [page 584](#).

Vomiting

In the initial phase of vomiting, the lower oesophageal sphincter and peri-oesophageal crural fibres simultaneously relax and rapid diaphragmatic and abdominal muscular contraction commences, increasing intra-abdominal pressure.

OESOPHAGEAL DYSMOTILITY

Oesophageal dysmotility encompasses disorders of the upper and lower oesophageal sphincters (discussed in detail in the section on gastro-oesophageal reflux), congenital anomalies such as atresia, and achalasia. Upper oesophageal sphincter dysfunction is usually neurological, often brainstem disease impairing relaxation of cricopharynx with resultant dysphagia and aspiration as demonstrated by videofluoroscopy.

Achalasia of the cardia is a primary motor disorder of the oesophagus in which there is failure of relaxation of the cardiac-oesophageal sphincter and loss of peristalsis in the oesophageal body from degeneration of myenteric plexus neurones. Dysphagia, regurgitation of undigested food and retrosternal chest pain result and a posteroanterior chest radiograph may show dilation of the oesophagus and retention of food products as a mottled, double-contoured mediastinal widening. Barium swallow shows a classic 'bird-beak' appearance as a result of failure of relaxation of the lower oesophageal sphincter and an absence of peristalsis. Oesophageal pressure manometry demonstrates absent or impaired relaxation of the lower oesophageal sphincter. Treatment options are endoscopic pneumatic balloon dilation or laparoscopic cardiomyotomy (Heller's myotomy). Alternatively, intrasphincteric injection of botulinum toxin may be offered in frail patients and confers lower morbidity and mortality.

Diffuse oesophageal spasm may present with dysphagia and chest pain. It is characterized by abnormal oesophageal contractions on barium swallow, and produces simultaneous segmental contractions that obliterate the oesophageal lumen to produce a 'corkscrew oesophagus'. Manometry also demonstrates these tetanic contractions. Non-specific oesophageal motility disorder occurs with ageing (presby-oesophagus) and is characterized by a decreased incidence of normal or even absent peristalsis after swallowing, often accompanied by repetitive tertiary contractions and failure of relaxation of the lower oesophageal sphincter. Oesophageal dysmotility is also seen in scleroderma and

Oesophagoscopy, a component of superior digestive endoscopic examination, may be indicated in patients with persistent non-cardiac chest pain, dysphagia, odynophagia (painful swallowing) or symptoms of reflux that include nocturnal cough. The endoscope is passed orally under direct vision with the patient in a lateral decubitus position, with or without sedation, and manoeuvred into the oesophageal inlet when the patient is asked to swallow; once through, the endoscope is advanced slowly under direct vision. The mucosa normally appears whitish pink, changing to red at the squamocolumnar junction (Z line) near the diaphragmatic hiatus. Cranial displacement of the Z line by more than 2 cm is indicative of a hiatus hernia. There may be slight extrinsic compression by the aorta and the left principal bronchus.

Endoscopic transoesophageal ultrasound permits assessment of the oesophageal wall and para-oesophageal structures, such as lymph nodes and the heart and aorta. Numerous surgical and percutaneous cardiovascular procedures are guided by transoesophageal echography. The technique is more sensitive than CT in detecting mediastinal nodal metastases, allowing those less than 10 mm in size to be assessed. Normal lymph nodes cannot be identified easily by endoscopic ultrasound because they have the same echogenicity characteristics as surrounding tissue; a hypoechoic or inhomogeneous echo portrayal on ultrasound is considered more suspicious of malignancy. Endoscopic ultrasound-guided fine needle aspiration allows accurate diagnosis for para-oesophageal nodes. Transbronchial fine needle aspiration is an alternative approach for sampling mediastinal nodes, guided by CT or positron emission tomography (PET) CT findings.

Oesophageal rupture

The majority of oesophageal ruptures are iatrogenic, and occur during the instrumentation of the oesophagus during simple endoscopy, endoscopic intervention including dilation, and nasogastric tube placement. Other causes are barotrauma, foreign bodies and oesophageal carcinoma. Barotrauma occurs with intense retching and vomiting that usually cause longitudinal tearing in the posterior wall of the thoracic oesophagus (Boerhaave's syndrome). Overall, locations of oesophageal perforation are 60% thoracic, 25% cervical and 15% abdominal. Perforation in the cervical oesophagus occurs in the posterior wall, where

it is at its thinnest, often corresponding to an area of mucosa with no muscular covering bordered by the pharyngeal inferior constrictor and cricopharyngeus. Mortality from oesophageal perforations is 20% and results from mediastinitis and necrotizing infection. Presenting symptoms include thoracic pain, odynophagia, haematemesis, subacute surgical emphysema, tachycardia, breathlessness and, in advanced cases, septic shock. Diagnosis and repair within 24 hours is associated with a better prognosis and a mortality rate of between 10% and 15%, whereas late diagnosis has a much greater mortality rate. For a perforation in the upper two-thirds of the thoracic oesophagus, a right thoracotomy in the fifth and sixth intercostal space is performed. The lower third of the oesophagus is approached through a left thoracotomy in the sixth and seventh intercostal space. The perforation is closed in layers after any necrotic tissue has been debrided.

A double-contrast barium swallow with fluoroscopy enables anatomical and functional assessment of the oesophagus (see Fig. 56.18). A standard examination involves the patient swallowing a barium solution and gas granules, and images are taken in the erect, supine and prone recumbent positions. The upright posture fully dilates the oesophagus and is useful in evaluating any constrictions. (The narrow tubular lumen of the oesophagus is normally indented slightly by the aortic arch and the left principal bronchus.) The supine view allows better evaluation of any motor dysfunction and varices, and moving into the prone position provokes underlying gastro-oesophageal reflux. Both repeating single digital images and cine loops of fluoroscopy can be obtained. The fluoroscopic images demonstrate defective opening of the pharyngeal oesophageal sphincter and the nature of the contractions in the body of the oesophagus. The cardio-oesophageal sphincter can be assessed in achalasia. Other abnormalities, such as hiatus hernia and areas of narrowing or dilation of the oesophagus, may be demonstrated. Mucosal relief images are obtained once the bulk of the contrast media has passed into the stomach, and this provides fine mucosal detail. Cross-sectional imaging is usually used for the preoperative planning and staging of oesophageal tumours, and is of value in radiotherapy planning.

other connective tissue diseases where atrophic smooth muscle is replaced by fibrous tissue in the submucosa and lamina propria.

GASTRO-OESOPHAGEAL REFLUX

Gastro-oesophageal reflux is described in [Chapter 64](#).

HIATUS HERNIA

Hiatus hernia is described in [Chapter 55](#).

MEDIASTINAL IMAGING

Mediastinal structures may be viewed using X-rays, CT and MRI (see [Commentary 7.1](#)).

NORMAL MEDIASTINAL CONTOURS ON A FRONTAL CHEST RADIOGRAPH

In the posteroanterior (PA) chest radiograph, the X-ray beam enters the back and exits the anterior chest wall during a full inspiratory breath-hold. The patient stands and elevates and abducts the arms over the radiographic detector; this protracts the scapulae and depresses the medial ends of the clavicles to clear the peripheral lung fields and apices, respectively. In this view, the heart and large blood vessels appear as the 'mediastinal silhouette' ([Fig. 56.16](#)). Forming its left border, superior to inferior, are the aortic arch ('aortic knuckle', sometimes with the left superior intercostal vein visible – the 'aortic nipple'), left pulmonary artery and left ventricle. The descending aorta is projected as a continuous left paravertebral border to the diaphragm. The right border is delineated by structures containing deoxygenated blood: superior to inferior, the right brachiocephalic vein, superior vena cava, right atrium and thoracic inferior vena cava. The right paratracheal stripe (up to 3 mm) is seen with the oval silhouette of the azygos arch at its lower margin.

Dilation or displacement of any of these structures accentuates the normal bulges on the borders of the mediastinal silhouette; thus an enlarged left auricle is seen between the left principal pulmonary artery and left ventricle as in mitral valve stenosis. Pulmonary vessels associated with the lung roots form the hilar contours. Sometimes seen are the anterior junctional line, representing supracardiac lung apposition (projects up into the neck because of the obliquity of the superior thoracic aperture), and inferiorly, the posterior junctional line, representing pleural apposition posterior to the oesophagus. The azygo-oesophageal line is sometimes seen, representing the medial border of the right lung in the azygo-oesophageal recess. The paraspinal lines are also visible parallel to the right and left margins of the thoracic spine.



Fig. 56.16 A posteroanterior (PA) radiograph of the adult female chest. Key: 1, trachea; 2, azygos vein; 3, right atrial border; 4, aortic arch; 5, left ventricular border.

NORMAL MEDIASTINAL CONTOURS ON A LATERAL CHEST RADIOGRAPH

On a lateral view, the cardiac silhouette is seen above the anterior, horizontal part of the diaphragm ([Fig. 56.17](#)). Due to the nomenclature of the cardiac chambers reflecting their prerotational embryonic state, the right ventricle and left atrium form the anterior and posterior cardiac contours, respectively. Posteriorly, both the descending thoracic aorta and the oesophagus in the posterior mediastinum may be seen; the oesophagus can be visualized with swallowed air or barium on a chest radiograph or fluoroscopically in the barium swallow examination ([Fig. 56.18](#)). The compressive effect on the oesophagus of a dilated left atrium will be readily identified during barium swallow. Superiorly, the less dense trachea and main bronchi are recognizable, the left principal bronchus appearing almost circular, seen end-on due to its more

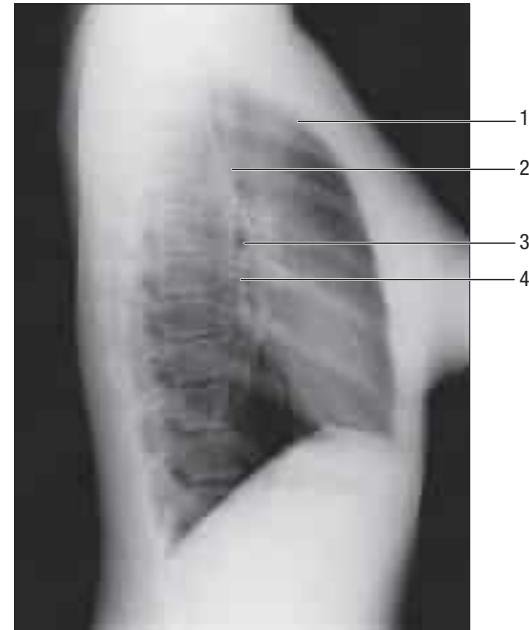


Fig. 56.17 A lateral radiograph of the adult female chest. Key: 1, retrosternal line (fat); 2, posterior tracheal line; 3, right principal bronchus; 4, bronchus intermedius.

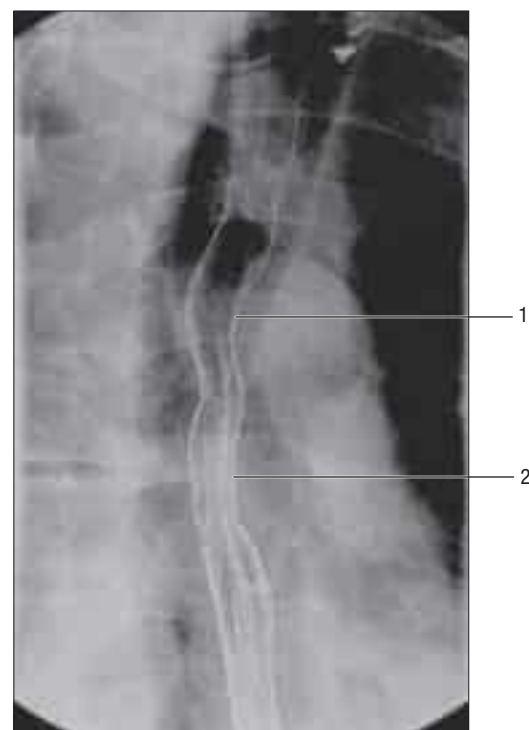


Fig. 56.18 An oblique radiograph of the thorax during barium swallow. Key: 1, indentation from aortic arch; 2, indentation from left principal bronchus.

An accessory lung fissure, the azygos fissure, is rarely present and seen in 0.5% of PA chest radiographs, extending from the lung apex to the right tracheobronchial angle. The azygos arch width should be no more than 5 mm, but may measure up to 10 mm in the third trimester of pregnancy as a result of a physiological blood volume increase and the fact that flow through the inferior vena cava may be diminished as a result of compression by the uterus, with compensatory drainage through the azygos system. Similarly, congenital inferior vena cava interruption at the diaphragm or acquired inferior vena cava thrombosis enlarges the azygos vein.

horizontal lie. The oesophagus is directly posterior to the trachea, and the adjacent anterior oesophageal and posterior tracheal walls appear as the posterior paratracheal line, seen in the majority of lateral radiographs (normally less than 5 mm).

The aortic arch and its main branches, anterior to the trachea, produce faint shadows in the superior mediastinum. The brachiocephalic veins are visible as an extrapleural bulge directly posterior to the manubrium. Retrosternal lung should always be seen; if it is absent, this raises a strong suspicion of a superior mediastinal tumour or

lymphadenopathy. The thoracic inferior vena cava is visible in the majority of patients as it drains into the right atrium.

CT AND MRI OF THE MEDIASTINAL SPACES

CT provides excellent mediastinal delineation (Figs 56.19–56.20) with intravenous contrast enhancement more readily distinguishing normal anatomy from pathological lymph nodes and tumours. The hilar,

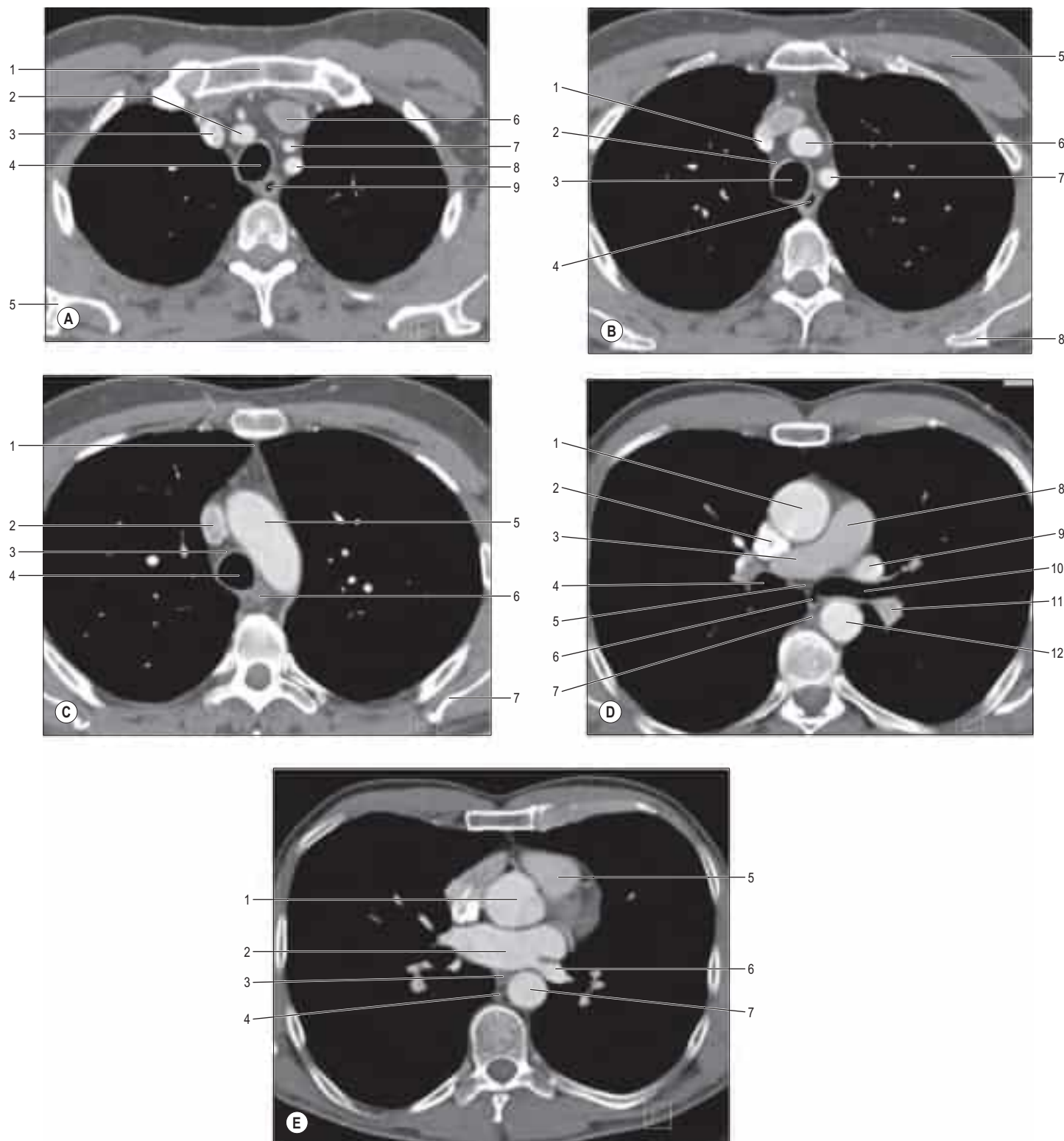


Fig. 56.19 Axial CT thoracic sections. **A**, At the level of the first costochondral junction. Key: 1, manubrium; 2, right brachiocephalic trunk; 3, right brachiocephalic vein; 4, trachea; 5, scapula; 6, left brachiocephalic vein; 7, left common carotid artery; 8, left subclavian artery; 9, oesophagus. **B**, At T3 level. Key: 1, brachiocephalic vein; 2, pretracheal space; 3, trachea; 4, oesophagus; 5, pectoralis major; 6, left subclavian artery arising from brachiocephalic trunk (normal variant); 7, left subclavian artery; 8, scapula. **C**, At T4 level. Key: 1, anterior junction; 2, superior vena cava; 3, pretracheal space; 4, trachea; 5, arch of aorta; 6, oesophagus; 7, scapula. **D**, At T6 level. Key: 1, ascending aorta; 2, superior vena cava; 3, right pulmonary artery; 4, right principal bronchus; 5, subcarinal space; 6, oesophagus; 7, azygos vein; 8, trunk of pulmonary artery; 9, left superior pulmonary vein; 10, left principal bronchus; 11, inferior branch of the left pulmonary artery; 12, descending aorta. **E**, At T7 level. Key: 1, aortic root; 2, left atrium; 3, oesophagus; 4, azygos vein; 5, right ventricular outflow tract; 6, left inferior pulmonary vein; 7, descending aorta.

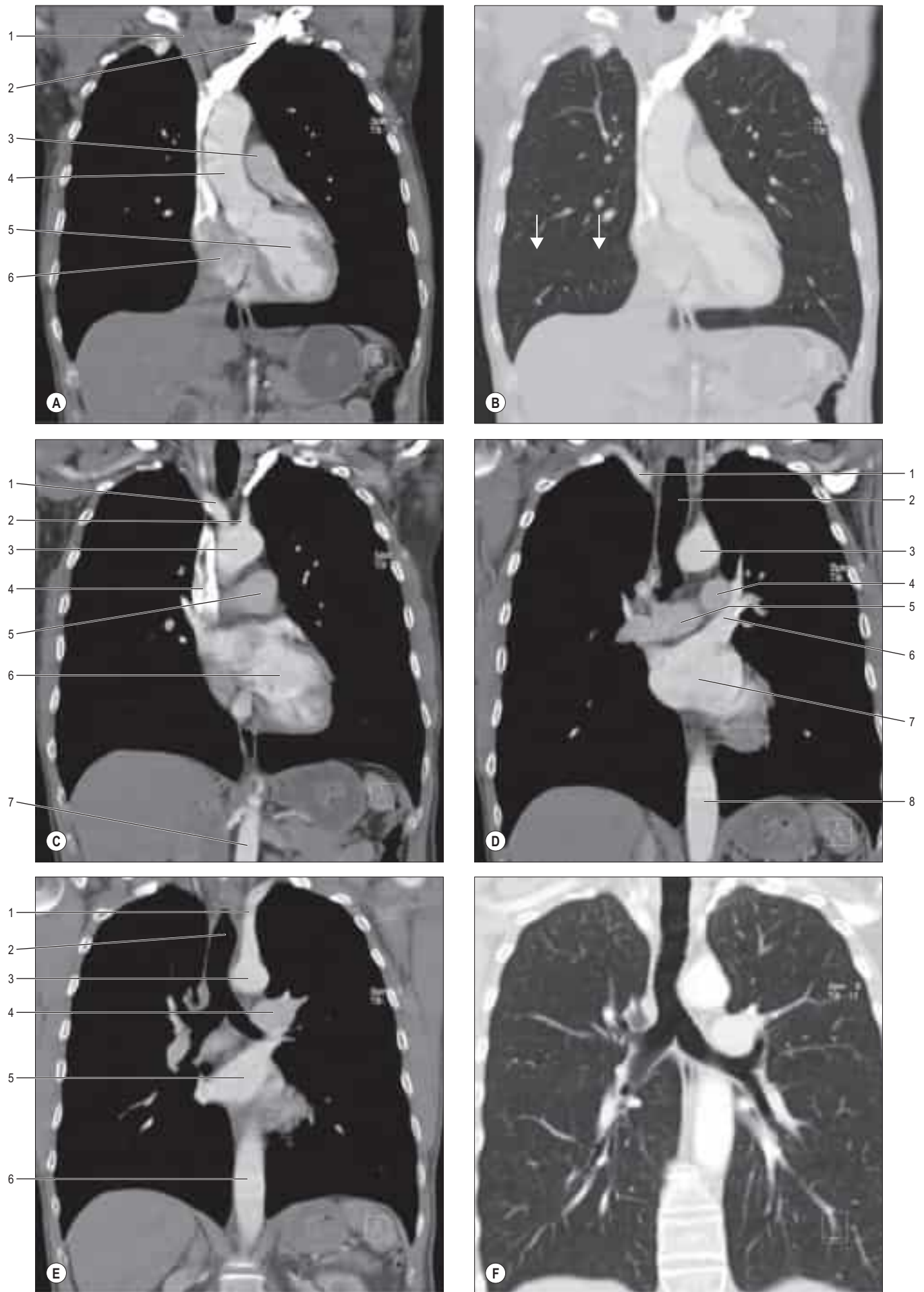


Fig. 56.20 Serial coronal thoracic CT sections. **A**, Key: 1, right brachiocephalic vein; 2, left brachiocephalic vein; 3, pulmonary trunk; 4, ascending aorta; 5, left ventricle; 6, right atrium. **B**, As **A**, but with lung window settings; horizontal (minor) fissure arrowed. **C**, Key: 1, brachiocephalic trunk; 2, left common carotid artery; 3, aortic arch; 4, superior vena cava; 5, pulmonary trunk; 6, left ventricle; 7, abdominal aorta. **D**, Key: 1, right subclavian artery; 2, trachea; 3, aortic arch; 4, left main pulmonary artery; 5, right main pulmonary artery; 6, left superior pulmonary vein; 7, left atrium; 8, descending thoracic aorta. **E**, Key: 1, left subclavian artery; 2, trachea; 3, posterior aortic arch; 4, left main pulmonary artery; 5, left atrium; 6, descending thoracic aorta. **F**, As **E**, but with lung window settings.

pretracheal, subcarinal, right paratracheal, posterior tracheal and aortopulmonary window stations are common sites of lymph node metastases in lung cancer.

The pretracheal space, as seen on CT, is bordered by the trachea posteriorly, the superior vena cava and right brachiocephalic vein anteriorly, and the descending aorta and superior pericardial sinus to the left. The aortopulmonary window is located between the aortic arch superiorly and the left pulmonary artery inferiorly; the trachea forms its medial border and the left lung its lateral border. The right paratracheal space lies between the right lung and the trachea on its anterolateral aspect; the posterior tracheal space between the lung and the posterolateral aspect of the trachea; and the subcarinal space inferior to the carina bounded by the principal bronchi. On the right, the azygo-oesophageal recess is located posterior to the subcarinal space

and on the left side is the oesophagus. All these areas are in direct continuity with each other and are inspected during transcervical mediastinoscopy.

The junctional areas are where the two lungs approach each other. The anterior junction lies between the great vessels, posterior margin of the chest wall and the lungs where the left brachiocephalic vein, highest mediastinal nodes, thymus and phrenic nerves are located. The posterior junction is an area posterior to the trachea and is where the lungs appose. The paraspinal area lies between the lateral margins of the spine and the lungs where the intercostal vessels, the ganglionated sympathetic chain and small lymph nodes are located. The retrocrural space, between the diaphragmatic crura and vertebral bodies, is traversed by the aorta, azygos venous system, thoracic duct, intercostal arteries, sympathetic chains and splanchnic nerves.

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